Theory of Linear Model Notes

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Contents

good luck :)

1 Stochastic Convergence

Convergence in Probability. X_n converges to X in probability if, for every $\epsilon > 0$,

$$\lim_{n\to\infty} \mathbb{P}\{|X_n - X| > \epsilon\} = 0$$

Almost Sure Convergence. X_n converges to X almost surely if

$$\mathbb{P}\{\lim_{n\to\infty} X_n = X\} = 1$$

Convergence in Distribution. X_n converges to X in distribution if

$$\lim_{n\to\infty} F_n(x) = F(x)$$

for every continuity point x of F, where F is the cdf of X.
Continuous Mapping Theorem. If $X_n \to X$ (in probability, in distribution, or a.s.) and g is continuous, then

$$g(X_n) \to g(X)$$

Weak Law of Large Numbers. Let $X_1, \dots, X_n$ be i.i.d. with $\mathbb{E}[X_i] = \mu$ and $\mathrm{Var}(X_i) = \sigma^2 < \infty$. Then

$$\bar{X}_n \xrightarrow{p} \mu$$

Strong Law of Large Numbers. Let $X_1, \dots, X_n$ be i.i.d. with $\mathbb{E}[X_i] = \mu$ and $\mathrm{Var}(X_i) = \sigma^2 < \infty$. Then

$$\bar{X}_n \xrightarrow{a.s.} \mu$$

Central Limit Theorem. Let $X_1, \dots, X_n$ be i.i.d. with $\mathbb{E}[X_i] = \mu$ and $\mathrm{Var}(X_i) = \sigma^2 < \infty$. Then

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} \mathrm{N}(0, \sigma^2)$$

Slutsky's Theorem. If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{p} a$, a constant, then

$$(1)\ Y_n X_n \xrightarrow{d} aX \qquad (2)\ X_n + Y_n \xrightarrow{d} X + a$$

Delta Method. Let $\sqrt{n}(X_n - \theta) \xrightarrow{d} \mathrm{N}(0, \sigma^2)$. If $g'(\theta)$ exists and is nonzero, then

$$\sqrt{n}\big[g(X_n) - g(\theta)\big] \xrightarrow{d} \mathrm{N}(0,\, \sigma^2[g'(\theta)]^2)$$

Taylor Series Approximation.
First-Order:

$$g(X_n) = g(\mu) + g'(\mu)(X_n - \mu) + \text{Remainder}$$

Second-Order:

$$g(X_n) = g(\mu) + g'(\mu)(X_n - \mu) + \frac{g''(\mu)}{2}(X_n - \mu)^2 + \text{Remainder}$$

Non-identical distribution CLT (multivariate) The multivariate form of the Lindeberg condition is considerably easier to state in "mean" form, so this is the form in which almost all textbooks present it.
Theorem (Lindeberg–Feller CLT). Suppose $\{X_{ni}\}$ is a triangular array of $d \times 1$ random vectors such that

$$z_n = \frac{1}{n}\sum_{i=1}^n X_{ni}, \quad V_n = \frac{1}{n}\sum_{i=1}^n \mathrm{Var}(X_{ni}) \longrightarrow V,$$

where V is positive definite. If for every $\epsilon > 0$,

$$\frac{1}{n}\sum_{i=1}^n \mathbb{E}\big[\|X_{ni}\|^2\, \mathbf{1}\big(\|X_{ni}\| \geq \epsilon\sqrt{n}\big)\big] \longrightarrow 0,$$

then

$$\sqrt{n}\, z_n \xrightarrow{d} \mathcal{N}(0, V).$$

Or equivalently,

$$\sqrt{n}\, V_n^{-1/2} z_n \xrightarrow{d} \mathcal{N}(0, I).$$

2 Matrix algebra

Singular Value Decomposition (SVD).
Full SVD. For $X_{n\times d}$ with $n \geq d$, there exist:

- $U_{n\times d} = (u_1, \ldots, u_d)$ with orthonormal columns: $U^\top U = I_d$,
- $V_{d\times d} = (v_1, \ldots, v_d)$ orthogonal: $V^\top V = VV^\top = I_d$,
- $\Sigma_{d\times d} = \mathrm{diag}(\sigma_1, \ldots, \sigma_d)$ with $\sigma_i \geq 0$,

such that

$$X_{n\times d} \;=\; U_{n\times d}\, \Sigma_{d\times d}\, V_{d\times d}^\top \;=\; \sum_{i=1}^{d} \sigma_i\, u_i v_i^\top.$$

The σ_i are the *singular values* of X; the columns of U and V are the *left* and *right singular vectors*, respectively.
<u>*Remark.*</u> *(Specrtral decompostion).* The Specrtral decompostion is around symmetric matrices, s.t.

$$X_{n\times n} \;=\; U_{n\times n}\, \Sigma_{n\times n}\, U_{n\times n}^\top$$

Thin (compact) SVD — when $\mathrm{rank}(X) = r < d$. If X has rank $r < d$, only r singular values are nonzero. We can write the *thin SVD*:

- $U_{n\times r} = (u_1, \ldots, u_r)$ — left singular vectors, $U^\top U = I_r$,
- $V_{d\times r} = (v_1, \ldots, v_r)$ — right singular vectors, $V^\top V = I_r$,
- $\Sigma_{r\times r} = \mathrm{diag}(\sigma_1, \ldots, \sigma_r)$ with $\sigma_i > 0$,

such that

$$X \;=\; U_{n\times r}\, \Sigma_{r\times r}\, V_{r\times d}^\top \;=\; \sum_{i=1}^{r} \sigma_i\, u_i v_i^\top.$$

The columns of U span $\mathcal{C}(X)$ (the column space), and the columns of V span $\mathcal{C}(X^\top)$ (the row space).
Key consequences.

i) $X^\top X = V\Sigma^2 V^\top$ — eigendecomposition of the Gram matrix.
ii) $XX^\top = U\Sigma^2 U^\top$ — same nonzero eigenvalues as $X^\top X$.

<u>*Remark.*</u> *(Orthogonality conventions).* In the full SVD, $V_{d\times d}$ is a square orthogonal matrix, i.e. $V^\top V = VV^\top = I_d$. In the thin SVD, $V_{d\times r}$ with $r < d$ has orthonormal columns ($V^\top V = I_r$) but is *not* square, so $VV^\top \neq I_d$. Instead, $VV^\top$ and $V_\perp V_\perp^\top$ are complementary orthogonal projection matrices satisfying $VV^\top + V_\perp V_\perp^\top = I_d$.

3 Projection

Projection of random variable $\hat{S}$ is the projection of T onto $\mathcal{S}$ iff

i) $\hat{S} \in \mathcal{S}$, and

ii) $\mathbb{E}\big[(T - \hat{S})\, S\big] = 0$, i.e. $\langle T - \hat{S},\, S \rangle = 0 \quad \forall\, S \in \mathcal{S}$

This projection is unique (in the sense that if $\hat{S}'$ is another projection then $\mathbb{P}\big\{\hat{S} \neq \hat{S}'\big\} = 0$).
If $\mathcal{S}$ contains the constant functions, then

$$\mathbb{E}\big[\hat{S}\big] = \mathbb{E}\big[T\big] \quad \text{and} \quad \mathrm{Cov}(T - \hat{S},\, S) = 0 \quad \forall\, S \in \mathcal{S}.$$

4 Regression

Model free approach

$$Y = \underbrace{\mathbb{E}\big[Y \mid X\big]}_{\text{best approx. of } Y \text{ using } X} + \underbrace{\big(Y - \mathbb{E}\big[Y \mid X\big]\big)}_{\varepsilon \text{ (error)}}$$

The function $x \in \mathbb{R}^d \mapsto \mathbb{E}\big[Y \mid X = x\big]$ is the regression function. So letting $f^{\text{opt}}(x) = \mathbb{E}\big[Y \mid X\big]$,

$$Y = \underbrace{f^{\text{opt}}(X)}_{\text{signal}} + \underbrace{\varepsilon}_{\text{error}}$$

Properties:
i) $\mathbb{E}\big[\varepsilon\big] = \mathbb{E}\big[Y - \mathbb{E}\big[Y \mid X\big]\big] = 0$
ii) $\mathbb{E}\big[f^{\text{opt}}(X)\,\varepsilon\big] = \text{Cov}\big[f^{\text{opt}}(X),\, \varepsilon\big] = 0$ (by orthogonality)

Signal + independent:

$$Y = \underbrace{f(X)}_{\text{smooth}} + \varepsilon, \quad \mathbb{E}\big[\varepsilon\big] = 0, \quad X \perp\!\!\!\perp \varepsilon$$

Parametric regression:

$$Y = \underbrace{f(X)}_{\text{parametric}} + \varepsilon, \quad \mathbb{E}\big[\varepsilon\big] = 0, \quad X \perp\!\!\!\perp \varepsilon$$

where f belongs to a parametric class, i.e. each f is parametrized by $\theta \in \Theta \subseteq \mathbb{R}^m$. So the regression function is f_{θ^*} for some $\theta^* \in \Theta$.

Linear regression:

$$Y = X^T \beta^* + \varepsilon, \quad \mathbb{E}\big[\varepsilon\big] = 0, \quad X \perp\!\!\!\perp \varepsilon$$

5 Linear Regression

Inference If the regression function is not necessarily linear, there still exists a natural target parameter — the β^* that gives the best linear approximation to the regression function. This is called the *projection parameter*. It is well defined provided that $\mathbb{E}\big[Y^2\big] < \infty$ and $\mathbb{E}\big[XX^\top\big]$ exists (is invertible). It is equal to

$$\begin{aligned}\beta^* &= \operatorname*{argmin}_{\beta\in\mathbb{R}^d} \mathbb{E}\Big[\big(\mathbb{E}[Y \mid X] - X^\top\beta\big)^2\Big] = \operatorname*{argmin}_{\beta\in\mathbb{R}^d} \mathbb{E}\Big[\big(Y - X^\top\beta\big)^2\Big] \\ &= \underbrace{\big(\mathbb{E}[XX^\top]\big)^{-1}}_{d\times d}\ \underbrace{\mathbb{E}[Y\cdot X]}_{d\times 1}\end{aligned}$$

Prediction

$$\begin{aligned}R(\beta) &= \mathbb{E}\Big[\big(Y_{\text{new}} - X_{\text{new}}^\top\beta\big)^2\Big] \\ &= \underbrace{\|\beta-\beta^*\|_\Sigma^2}_{\text{estimation error}} + \underbrace{\sigma^2}_{\text{variance}} + \underbrace{\eta^2}_{\text{non-linearity}}\end{aligned}$$

where $\sigma^2 = \mathbb{E}\Big[\big(Y - \mathbb{E}[Y \mid X]\big)^2\Big]$, $\eta^2 = \mathbb{E}\Big[\big(\mathbb{E}[Y \mid X] - X^\top\beta^*\big)^2\Big]$, and $\beta^* = \big(\mathbb{E}[XX^\top]\big)^{-1}\mathbb{E}[Y\cdot X]$ is the projection parameter.
Case 1: If $\beta = \beta^*$, then $R(\beta^*) = \sigma^2 + \eta^2 = \inf_{\beta\in\mathbb{R}^d} R(\beta)$.
Case 2: If linearity holds, i.e. $Y = X^\top\beta^* + \varepsilon$, $\mathbb{E}[\varepsilon] = 0$, $\varepsilon \perp\!\!\!\perp X$, then $\eta = 0$ and

$$R(\beta^*) = \sigma^2 = \mathbb{E}\Big[\big(Y - X^\top\beta^*\big)^2\Big] = \mathbb{E}\big[\varepsilon^2\big] = \mathrm{Var}(\varepsilon).$$

Estimation We need to estimate β^* (either the projection parameter, or the actual parameter if the model is linear). The plug-in estimator of β^* is

$$\begin{aligned}\hat\beta &= \operatorname*{argmin}_{\beta\in\mathbb{R}^d} \hat R(\beta) \\ &= (X^TX)^{-1}X^TY \\ &= \big(\hat{\mathbb{E}}_n\big[X_iX_i^\top\big]\big)^{-1}\hat{\mathbb{E}}_n\big[Y_i\cdot X_i\big]\end{aligned}$$

where the empirical MSE (risk) is

$$\begin{aligned}\hat R(\beta) &= \hat{\mathbb{E}}_n\Big[\big(Y_i - X_i^\top\beta\big)^2\Big] \\ &= \frac{1}{n}\sum_{i=1}^n \big(Y_i - X_i^\top\beta\big)^2 \\ &= \frac{1}{n}\Big\|\underbrace{Y}_{n\times 1} - \underbrace{X}_{n\times d}\beta\Big\|^2\end{aligned}$$

Geometric Interpretation

$$\hat Y = X\hat\beta = \underbrace{X(X^\top X)^{-1}X^\top}_{H}\, Y = HY$$

where H is the hat matrix, the orthogonal projection onto $\mathcal{C}(X)$ (H is symmetric and $H^2 = H\cdot H = H$). The residuals are

$$e = Y - \hat Y = \underbrace{(I_n - H)}_{\text{proj. onto } \mathcal{C}(X)^\perp}\, Y \in \mathbb{R}^n$$

where $I_n - H$ is also an orthogonal projection (symmetric & idempotent) onto $\mathcal{C}(X)^\perp$.
Remarks
Remark 1. If matrix $P \in \mathbb{R}^{n\times n}$ is a projection matrix, then $\mathrm{trace}(P) = \mathrm{rank}(P) = \dim(\mathrm{col}(P)) = \dim(\text{target space})$. Since the projection matrix P always project $\forall y \in \mathbb{R}^n$ onto the column space of P.
Remark 2. $\mathrm{trace}(X(X^TX)^{-1}X^T) = \mathrm{rank}(X(X^TX)^{-1}X^T) = \dim(\mathrm{col}(X(X^TX)^{-1}X^T)) = \dim(\mathrm{col}(X))$.

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Min-norm OLS estimator. When $\mathrm{rank}(X) < d$, the normal equations $X^\top X\hat{\beta} = X^\top Y$ have infinitely many solutions. Among these, there is a canonical one: the *min-norm solution*, the one with the smallest Euclidean norm. It is defined as

$$\hat{\beta}_{\mathrm{MN}} \;=\; (X^\top X)^{+}X^\top Y \;=\; X^{+}Y,$$

where X^{+} denotes the Moore–Penrose pseudo-inverse of X.
Remark. (Equivalence of the two forms via SVD). Let $X = U\Sigma V^\top$ be the thin SVD of X with $\mathrm{rank}(X) = r$. Then $X^\top X = V\Sigma^2 V^\top$, so $(X^\top X)^{+} = V\Sigma^{-2}V^\top$. Therefore:

$$(X^\top X)^{+}X^\top \;=\; V\Sigma^{-2}V^\top \cdot V\Sigma U^\top \;=\; V\Sigma^{-1}U^\top \;=\; X^{+},$$

confirming $(X^\top X)^{+}X^\top = X^{+}$ and hence both expressions for $\hat{\beta}_{\mathrm{MN}}$ are identical. Note: when connecting $\hat{\beta}_{\mathrm{MN}}$ to the ridge estimator $\hat{\beta}_\lambda$ in the limit $\lambda \downarrow 0$, we use the form $\hat{\beta}_{\mathrm{MN}} = X^{+}Y$.
Moore–Penrose pseudo-inverse. For a matrix $A_{m\times n}$, its Moore–Penrose pseudo-inverse A^{+} is the unique $n \times m$ matrix satisfying:

i) $AA^{+}A = A$ (AA^{+} maps columns of A to themselves, i.e. it acts as the identity on $\mathcal{C}(A)$),
ii) $A^{+}AA^{+} = A^{+}$,
iii) AA^{+} and $A^{+}A$ are symmetric,

Pseudo-inverse via SVD. Let $A_{m\times n} = U_{m\times k}\,\Sigma_{k\times k}\,V_{k\times n}^\top$ be the thin SVD of A, where $\Sigma = \mathrm{diag}(\sigma_1, \ldots, \sigma_k)$ with $\sigma_i > 0$ and $k = \mathrm{rank}(A)$. Then:

$$A^{+} \;=\; V\,\Sigma^{-1}\,U^\top.$$

Back to interpolation. When $\mathrm{rank}(X) = n$ (i.e. X has full row rank, which requires $d \geq n$), the min-norm estimator satisfies:

$$\hat{\beta}_{\mathrm{MN}} \;=\; X^{+}Y \;=\; \underset{\beta\in\mathbb{R}^d}{\mathrm{argmin}}\big\{\|\beta\| \;\text{ s.t. }\; X\beta = Y\big\}.$$

6 Gauss-Markov Theorem

Setup. Assume a linear model with *fixed* (deterministic) covariates $\Phi \in \mathbb{R}^{n\times d}$:

$$Y = \Phi\beta^* + \varepsilon, \qquad \varepsilon \sim \text{i.i.d.}\big(0, \sigma^2\big)$$

That is, Φ is deterministic and the errors are homoscedastic.
Linear unbiased estimators. Any estimator of β^* of the form $\tilde{\beta} = AY$ (with $A \in \mathbb{R}^{d\times n}$) is a *linear unbiased estimator* of β^* if

$$\mathbb{E}\big[AY\big] = \beta^* \qquad \text{for all } \beta^* \in \mathbb{R}^d.$$

Expanding: $\mathbb{E}\big[AY\big] = A\Phi\beta^*$. For this to equal β^* for *all* β^*, we need:

$$A\Phi = I_d.$$

In particular, the OLS estimator $\hat{\beta} = \big(\Phi^\top\Phi\big)^{-1}\Phi^\top Y$ is a linear unbiased estimator (set $A = \big(\Phi^\top\Phi\big)^{-1}\Phi^\top$, then $A\Phi = I_d$).
Gauss-Markov Theorem. Under the model above, $\hat{\beta}$ is the best linear unbiased estimator (**BLUE**). That is, for any other linear unbiased estimator $\tilde{\beta} = AY$,

$$\operatorname{Var}\big(\hat{\beta}\big) \;\preceq\; \operatorname{Var}\big(\tilde{\beta}\big).$$

<u>*Remark 1.*</u> *(The PSD order $\preceq$).* For two PSD matrices M_1, M_2 of the same size,

$$M_1 \preceq M_2 \qquad \Longleftrightarrow \qquad M_2 - M_1 \succeq 0 \quad \text{(positive semidefinite, psd order)}.$$

<u>*Remark 2.*</u> *(Coordinate-wise and contrast interpretation).* $\operatorname{Var}\big(\hat{\beta}\big) \preceq \operatorname{Var}\big(\tilde{\beta}\big)$ means that for any $z \in \mathbb{R}^d$,

$$z^\top \operatorname{Var}\big(\hat{\beta}\big)\, z \;\le\; z^\top \operatorname{Var}\big(\tilde{\beta}\big)\, z.$$

Choosing $z = e_i$ (the i-th standard basis vector) yields the coordinate-wise statement:

$$\operatorname{Var}\big(\hat{\beta}_i\big) \;\le\; \operatorname{Var}\big(\tilde{\beta}_i\big) \quad \text{for all } i.$$

In fact, the result is stronger: it holds for every *contrast* $c^\top\beta^*$ with $c \in \mathbb{R}^d$:

$$\operatorname{Var}\big(c^\top\hat{\beta}\big) \;\le\; \operatorname{Var}\big(c^\top\tilde{\beta}\big) \quad \text{for all } c \in \mathbb{R}^d.$$

Proof. Let $\tilde{\beta} = AY$ with $\mathbb{E}\big[\tilde{\beta}\big] = \beta^*$, so $A\Phi = I_d$. Since $\operatorname{Var}(Y) = \sigma^2 I_n$,

$$\operatorname{Var}\big(\tilde{\beta}\big) = A\operatorname{Var}(Y)\,A^\top = \sigma^2 AA^\top.$$

We want to show $\sigma^2 AA^\top \succeq \sigma^2\big(\Phi^\top\Phi\big)^{-1}$, equivalently:

$$AA^\top - \big(\Phi^\top\Phi\big)^{-1} \;\succeq\; 0.$$

Key construction. Define

$$D \;:=\; A - \big(\Phi^\top\Phi\big)^{-1}\Phi^\top.$$

Using $A\Phi = I_d$, this matrix satisfies:

$$D\Phi \;=\; A\Phi - \big(\Phi^\top\Phi\big)^{-1}\underbrace{\Phi^\top\Phi}_{I_d} \;=\; I_d - I_d \;=\; 0.$$

That is, D is orthogonal to the column space of Φ.
Expanding $AA^\top$. Substituting $A = \big(\Phi^\top\Phi\big)^{-1}\Phi^\top + D$ and expanding:

$$AA^\top \;=\; \underbrace{\big(\Phi^\top\Phi\big)^{-1}\Phi^\top\Phi\big(\Phi^\top\Phi\big)^{-1}}_{\left(\Phi^\top\Phi\right)^{-1}} + \underbrace{\big(\Phi^\top\Phi\big)^{-1}\overbrace{\Phi^\top D^\top}^{(D\Phi)^\top=0}}_{0} + \underbrace{\overbrace{D\Phi}^{=0}\big(\Phi^\top\Phi\big)^{-1}}_{0} + DD^\top.$$

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Both cross terms vanish because $D\Phi = 0$, leaving:

$$AA^\top \;=\; \left(\Phi^\top\Phi\right)^{-1} + DD^\top.$$

Therefore:

$$AA^\top - \left(\Phi^\top\Phi\right)^{-1} \;=\; DD^\top \;\succeq\; 0,$$

since $z^\top DD^\top z = \left\|D^\top z\right\|^2 \geq 0$ for all $z \in \mathbb{R}^d$. Hence $\mathrm{Var}\big(\tilde{\beta}\big) \succeq \mathrm{Var}\big(\hat{\beta}\big)$.

Remark 3. Why not prove the inequality by multiplying both sides of $AA^\top - \left(\Phi^\top\Phi\right)^{-1} \succeq 0$ by $\left(\Phi^\top\Phi\right)$ on one side? Unlike scalars, the matrix inequality $M \succeq 0$ does *not* imply $MC \succeq 0$ under one-sided multiplication, because MC need not be symmetric. Only a *congruence transformation* $PMP^\top$ preserves the PSD property. The D-matrix construction avoids this issue entirely.

Extension to random design. When Φ is a random matrix, the Gauss-Markov conclusion $\mathrm{Var}(\tilde{\beta}) \succeq \mathrm{Var}(\hat{\beta})$ still holds. The proof strategy is:

i) **Condition on Φ.** For fixed Φ, the result above applies directly: $\mathrm{Var}(\tilde{\beta} \mid \Phi) \succeq \mathrm{Var}(\hat{\beta} \mid \Phi) = \sigma^2(\Phi^\top\Phi)^{-1}$.

ii) **Take expectations via the Law of Total Variance.**

$$\mathrm{Var}\big(\tilde{\beta}\big) = \underbrace{\mathbb{E}\big[\mathrm{Var}\big(\tilde{\beta} \mid \Phi\big)\big]}_{\succeq\, \mathbb{E}\left[\mathrm{Var}(\hat{\beta}\mid\Phi)\right]} + \underbrace{\mathrm{Var}\big(\mathbb{E}\big[\tilde{\beta} \mid \Phi\big]\big)}_{\succeq\, 0} \;\succeq\; \mathbb{E}\big[\mathrm{Var}\big(\hat{\beta} \mid \Phi\big)\big].$$

iii) **Identify $\mathbb{E}\big[\mathrm{Var}\big(\hat{\beta} \mid \Phi\big)\big] = \mathrm{Var}\big(\hat{\beta}\big)$.** Apply the Law of Total Variance to $\hat{\beta}$ itself:

$$\mathrm{Var}\big(\hat{\beta}\big) = \mathbb{E}\big[\mathrm{Var}\big(\hat{\beta} \mid \Phi\big)\big] + \underbrace{\mathrm{Var}\big(\mathbb{E}\big[\hat{\beta} \mid \Phi\big]\big)}_{=\,\mathrm{Var}(\beta^*)\,=\,0}.$$

The second term vanishes because $\mathbb{E}\big[\hat{\beta} \mid \Phi\big] = \beta^*$ is a constant (not random). Therefore:

$$\mathbb{E}\big[\mathrm{Var}\big(\hat{\beta} \mid \Phi\big)\big] \;=\; \mathrm{Var}\big(\hat{\beta}\big) \;=\; \sigma^2\,\mathbb{E}\big[\left(\Phi^\top\Phi\right)^{-1}\big].$$

Combining with step (ii): $\mathrm{Var}\big(\tilde{\beta}\big) \succeq \mathrm{Var}\big(\hat{\beta}\big)$.

good luck :)

7 Ridge Regression

Motivation. When d/n approaches 1, we are essentially memorizing the observations y_i (that is, e.g., when $d = n$ and X is a square invertible matrix, $\hat{\beta} = X^{-1}y$ leads to $\hat{y} = X\hat{\beta} = y$; that is, OLS will lead to a perfect fit, which is typically not good for generalization to unseen data). Also, when $d > n$, $X^\top X$ is not invertible, and the normal equations admit a linear subspace of solutions. These behaviors of OLS in high dimensions (d large) are often undesirable.
One approach is to *regularize* — to solve a penalized least-squares problem that includes a penalty for not choosing "good" solutions.
Ridge estimator. For a regularization parameter $\lambda > 0$, the ridge estimator $\hat{\beta}_\lambda$ is defined as:

$$\hat{\beta}_\lambda \;=\; \operatorname*{argmin}_{\beta \in \mathbb{R}^d} \underbrace{\frac{\|Y - X\beta\|^2}{n} + \lambda\|\beta\|^2}_{f(\beta)}.$$

The objective function $f(\beta)$ is strictly convex (its Hessian is $\frac{2}{n}X^\top X + 2\lambda I_d \succ 0$), so the minimizer is unique. The first-order optimality conditions are:

$$0 \;=\; \nabla f(\hat{\beta}_\lambda) \;=\; \frac{2}{n}X^\top\big(X\hat{\beta}_\lambda - Y\big) + 2\lambda\hat{\beta}_\lambda.$$

Solving for $\hat{\beta}_\lambda$ gives the closed-form solution:

$$\hat{\beta}_\lambda \;=\; \big(\frac{X^\top X}{n} + \lambda I_d\big)^{-1}\frac{X^\top Y}{n} \;=\; \big(\hat{\Sigma} + \lambda I_d\big)^{-1}\frac{X^\top Y}{n},$$

where $\hat{\Sigma} = \frac{X^\top X}{n}$ is the empirical second moment matrix. This solution exists and is unique even if $d > n$ and regardless of how poorly conditioned $X^\top X$ is, since $\hat{\Sigma} + \lambda I_d \succ 0$ for all $\lambda > 0$.
Remarks
<u>Remark 1.</u> *(SVD interpretation).* Let $X = U\Sigma V^\top$ be the thin SVD of $X_{n\times d}$ with $\operatorname{rank}(X) = r$. Then the fitted values under Ridge are:

$$\hat{Y} \;=\; X\hat{\beta}_\lambda \;=\; \sum_{j=1}^{r} u_j\langle Y, u_j\rangle\,\frac{\sigma_j^2}{\sigma_j^2 + \lambda},$$

where u_j is the j-th column of U and σ_j is the j-th singular value of X. In contrast, the OLS fitted values are:

$$\hat{Y} \;=\; X\hat{\beta} \;=\; \sum_{j=1}^{r} u_j\langle Y, u_j\rangle.$$

Ridge shrinks each component by a factor $\frac{\sigma_j^2}{\sigma_j^2+\lambda} < 1$. Directions with small singular values (poorly determined by the data) are shrunk more aggressively.
<u>Remark 2.</u> *(Connection to min-norm OLS).* If $\lambda \to 0$, then $\hat{\beta}_\lambda \to \hat{\beta}_{\mathrm{MN}}$ (the min-norm estimator) or $\hat{\beta}$ (the OLS estimator) if $X^\top X$ is invertible.
<u>Remark 3.</u> *(Alternative expression and scaling convention).* By the push-through identity (HW 4, Q1), the ridge estimator admits an equivalent dual form:

$$\hat{\beta}_\lambda \;=\; \big(\frac{X^\top X}{n} + \lambda I_d\big)^{-1}\frac{X^\top Y}{n} \;=\; \frac{X^\top}{n}\big(\frac{XX^\top}{n} + \lambda I_n\big)^{-1}Y.$$

Some references omit the $1/n$ scaling and write instead:

$$\hat{\beta}_\lambda \;=\; \big(X^\top X + \lambda I_d\big)^{-1}X^\top Y \;=\; X^\top\big(XX^\top + \lambda I_n\big)^{-1}Y.$$

The two conventions are equivalent up to a rescaling of λ: replacing λ in the scaled version by $n\lambda$ recovers the unscaled version. The choice of convention does not affect the structure of the estimator.

Proof of Remark 1. We use the full SVD $X = U\Sigma V^\top$, where $U_{n\times n}$ and $V_{d\times d}$ are orthogonal matrices and $\Sigma_{n\times d}$ is diagonal with $\sigma_1 \geq \cdots \geq \sigma_r > 0 = \sigma_{r+1} = \cdots = \sigma_d$. We use the unscaled convention $\hat{\beta}_\lambda = \left(X^\top X + \lambda I_d\right)^{-1} X^\top Y$.

Ridge fitted values.

Step 1: Diagonalize $X^\top X + \lambda I_d$. Since V is orthogonal ($V^\top V = VV^\top = I_d$):

$$X^\top X = V\Sigma^\top \Sigma V^\top \quad \Rightarrow \quad X^\top X + \lambda I_d = V(\Sigma^\top\Sigma + \lambda I_d)V^\top,$$

where $\Sigma^\top\Sigma = \mathrm{diag}(\sigma_1^2, \ldots, \sigma_r^2, 0, \ldots, 0)$, so $\Sigma^\top\Sigma + \lambda I_d = \mathrm{diag}(\sigma_1^2+\lambda, \ldots, \sigma_r^2+\lambda, \lambda, \ldots, \lambda)$. Since all diagonal entries are strictly positive for any $\lambda > 0$, the inverse is:

$$\left(X^\top X + \lambda I_d\right)^{-1} = V(\Sigma^\top\Sigma + \lambda I_d)^{-1} V^\top.$$

Step 2: Compute $\left(X^\top X + \lambda I_d\right)^{-1} X^\top$.

$$\left(X^\top X + \lambda I_d\right)^{-1} X^\top = V(\Sigma^\top\Sigma + \lambda I_d)^{-1} \underbrace{V^\top V}_{I_d} \Sigma^\top U^\top = V(\Sigma^\top\Sigma + \lambda I_d)^{-1}\Sigma^\top U^\top.$$

Step 3: Compute $X\hat{\beta}_\lambda$.

$$\begin{aligned} X\hat{\beta}_\lambda &= U\Sigma \underbrace{V^\top V}_{I_d} (\Sigma^\top\Sigma + \lambda I_d)^{-1}\Sigma^\top U^\top Y \\ &= U\,\mathrm{diag}\Big(\frac{\sigma_1^2}{\sigma_1^2+\lambda}, \ldots, \frac{\sigma_r^2}{\sigma_r^2+\lambda}, 0, \ldots, 0\Big)\, U^\top Y, \end{aligned}$$

where the zero entries (for $j > r$) contribute nothing to the sum. Therefore:

$$\hat{Y} \;=\; X\hat{\beta}_\lambda \;=\; \sum_{j=1}^{r} u_j \langle Y,\, u_j\rangle\, \frac{\sigma_j^2}{\sigma_j^2+\lambda}.$$

OLS fitted values. The hat matrix using the full SVD is:

$$H = X(X^\top X)^{-1}X^\top = U\Sigma V^\top \cdot V(\Sigma^\top\Sigma)^{-1}V^\top \cdot V\Sigma^\top U^\top = U\,\mathrm{diag}(\underbrace{1, \ldots, 1}_{r}, \underbrace{0, \ldots, 0}_{d-r})\, U^\top = \sum_{j=1}^{r} u_j u_j^\top,$$

so $\hat{Y} = HY = \sum_{j=1}^{r} u_j \langle Y,\, u_j\rangle$.

Taking $\lambda \downarrow 0$ in the Ridge expression recovers the OLS expression, since $\frac{\sigma_j^2}{\sigma_j^2+\lambda} \to 1$.

Proof of Remark 2. We use the full SVD $X = U\Sigma V^\top$ throughout, where $U_{n\times n}$ and $V_{d\times d}$ are orthogonal and $\Sigma_{n\times d}$ is diagonal with $\sigma_1 \geq \cdots \geq \sigma_r > 0 = \sigma_{r+1} = \cdots$. We show that all three estimators produce the same fitted values $\sum_{j=1}^{r} u_j \langle Y,\, u_j\rangle$.

Case 1: OLS $\hat{\beta} = (X^\top X)^{-1}X^\top Y$ — when $X^\top X$ is invertible ($r = d$). When $r = d$, $\Sigma^\top\Sigma = \mathrm{diag}(\sigma_1^2, \ldots, \sigma_d^2)$ is invertible, so:

$$\begin{aligned} X\hat{\beta} &= X(X^\top X)^{-1}X^\top Y \\ &= U\Sigma \underbrace{V^\top V}_{I_d} (\Sigma^\top\Sigma)^{-1} \underbrace{V^\top V}_{I_d} \Sigma^\top U^\top Y \\ &= U \underbrace{\Sigma(\Sigma^\top\Sigma)^{-1}\Sigma^\top}_{\mathrm{diag}(1,\ldots,1)} U^\top Y \;=\; \sum_{j=1}^{r} u_j \langle Y,\, u_j\rangle. \end{aligned}$$

Case 2: Min-norm OLS $\hat{\beta}_{\mathrm{MN}} = X^+ Y$ — when $X^\top X$ is not invertible ($r < d$). The pseudo-inverse via SVD is $X^+ = V\Sigma^+ U^\top$, where $\Sigma^+_{d\times n}$ has entries $1/\sigma_j$ for $j \leq r$ and 0 otherwise. Then:

$$\begin{aligned} X\hat{\beta}_{\mathrm{MN}} &= XX^+Y = U\Sigma \underbrace{V^\top V}_{I_d} \Sigma^+ U^\top Y \\ &= U \underbrace{\Sigma\Sigma^+}_{\mathrm{diag}(1,\ldots,1,0,\ldots,0)} U^\top Y \;=\; \sum_{j=1}^{r} u_j \langle Y,\, u_j\rangle. \end{aligned}$$

Case 3: Ridge in the limit $\lim_{\lambda\downarrow 0}\hat{\beta}_\lambda$. From Remark 1:

$$\lim_{\lambda\downarrow 0} X\hat{\beta}_\lambda = \sum_{j=1}^{r} u_j\langle Y,\, u_j\rangle \underbrace{\lim_{\lambda\downarrow 0}\frac{\sigma_j^2}{\sigma_j^2+\lambda}}_{=1} = \sum_{j=1}^{r} u_j\langle Y,\, u_j\rangle.$$

All three cases give the same fitted values. In particular, $\hat{\beta}_{\mathrm{MN}} = \lim_{\lambda\downarrow 0}\hat{\beta}_\lambda$, and when $X^\top X$ is invertible, $\hat{\beta}_{\mathrm{MN}} = \hat{\beta}$.

Proof of Remark 3. We use the scaled convention. We want to show:

$$\Big(\frac{X^\top X}{n} + \lambda I_d\Big)^{-1}\frac{X^\top}{n} \;=\; \frac{X^\top}{n}\Big(\frac{XX^\top}{n} + \lambda I_n\Big)^{-1}.$$

Denote $A = \frac{X^\top X}{n} + \lambda I_d$ and $B = \frac{XX^\top}{n} + \lambda I_n$, so the claim is $A^{-1}\frac{X^\top}{n} = \frac{X^\top}{n}B^{-1}$. Left-multiplying both sides by A and right-multiplying by B:

$$A^{-1}\frac{X^\top}{n} = \frac{X^\top}{n}B^{-1} \qquad \Longleftrightarrow \qquad A\,\frac{X^\top}{n} = \frac{X^\top}{n}\,B.$$

We verify both sides are equal:

$$\begin{aligned} A\,\frac{X^\top}{n} &= \Big(\frac{X^\top X}{n} + \lambda I_d\Big)\frac{X^\top}{n} = \frac{X^\top XX^\top}{n^2} + \lambda\frac{X^\top}{n}, \\ \frac{X^\top}{n}\,B &= \frac{X^\top}{n}\Big(\frac{XX^\top}{n} + \lambda I_n\Big) = \frac{X^\top XX^\top}{n^2} + \lambda\frac{X^\top}{n}. \end{aligned}$$

Since $A\,\frac{X^\top}{n} = \frac{X^\top}{n}\,B$, the identity follows.

8 Risk decomposition

Risk overview

Let (X,Y) be a random pair in $\mathcal{X}\times\mathcal{Y}$. Training data $(X_1,Y_1),\ldots,(X_n,Y_n)$ are i.i.d. copies of (X,Y). Loss function $\ell:\mathcal{Y}\times\mathcal{Y}\to[0,\infty)$.
Risk: $R(h)=\mathbb{E}\big[\ell(h(X),Y)\big]$.
Empirical risk: $R_n(h)=\frac{1}{n}\sum_{i=1}^n \ell(h(X_i),Y_i)$.
$h^*=\operatorname{argmin}_{h\in\mathcal{H}} R(h),\quad \hat{h}_n=\operatorname{argmin}_{h\in\mathcal{H}} R_n(h)$.
Excess risk: $R(\hat{h}_n)-R(h^*)$.

Fixed design risk decomposition

Fixed design: $Y_i=X_i^\top\beta^*+\varepsilon_i$, where $\varepsilon_i\sim(0,\sigma^2)$ and $X_1,\ldots,X_n\in\mathbb{R}^d$ are fixed (deterministic) vectors in $\mathbb{R}^d$.
Two assumptions
1. Linearity. $E[y_i]=X_i^T\beta^*$. If $X_i's$ were random, linearity means $E[y_i\mid X_i]=X_i^T\beta^*$
2. Fixed covariates.

Risk in fixed design

Now, for **any** $\beta\in\mathbb{R}^d$. The risk of β is

$$\begin{aligned}R(\beta)&=\mathbb{E}_Y\big[\frac{\|Y-X\beta\|^2}{n}\big]=\mathbb{E}_\varepsilon\big[\frac{\|X(\beta^*-\beta)+\varepsilon\|^2}{n}\big]\\&=(\beta^*-\beta)^\top\frac{X^\top X}{n}(\beta^*-\beta)+\mathbb{E}_\varepsilon\big[\frac{\|\varepsilon\|^2}{n}\big]\\&=(\beta^*-\beta)^\top\hat{\Sigma}\,(\beta^*-\beta)+\sigma^2\\&=\big\|\beta^*-\beta\big\|_{\hat{\Sigma}}^2+\underbrace{\sigma^2}_{R(\beta^*)}\end{aligned}$$

The quantity $R(\beta)-R(\beta^*)=\big\|\beta^*-\beta\big\|_{\hat{\Sigma}}^2\geq 0$ is the **excess risk**.
Remarks
Remark 1. Based on the definition of risk function and fixed design linear regression, the loss function here is $\frac{\|Y-X\beta\|^2}{n}$ and the empirical risk function here is also same expression of loss function: $\frac{1}{n}\sum_{i=1}^n(Y_i-X_i^T\beta)^2$, which is same as random design's empirical risk.
Remark 2. You can think of $R(\beta)$ as $\mathbb{E}_{Y|X}\big[\frac{\|Y-X\beta\|^2}{n}\mid X\big]$ if X were random (assuming linearity). In the above risk decomposition formula, estimator β is constant with respect to ϵ. Because ϵ here is ϵ_{new}, while the randomness of β is caused by ϵ_{trian}. And $\epsilon_{new}\perp\!\!\!\perp\epsilon_{train}$.
Remark 3. You can think of $R(\beta)$ as $\mathbb{E}_{Y_{\text{new}}}\big[\frac{\|Y_{\text{new}}-X\beta\|^2}{n}\big]$ where $Y_{\text{new}}\in\mathbb{R}^n$ is a new draw of data independent of the observations. If you want to study the risk function for OLS estimator $\hat{\beta}$, let's $R(\hat{\beta})$, you should better know $R(\hat{\beta})$ is a r.v. And the randomness is caused by y_{train} / y or ϵ_{train} / ϵ.

Expected excess risk

The excess risk $R(\hat{\beta})-R(\beta^*)$ is a random variable. Thus, to analyze it, we take its expectation, where $\hat{\beta}$ denotes any data-driven estimator, such as the OLS estimator. Then, we can perform the bias-variance tradeoff for the expected excess risk of any estimator $\hat{\beta}$,

$$\mathbb{E}\big[\mathcal{R}(\hat{\beta})\big]-\mathcal{R}^*=\underbrace{\big\|\mathbb{E}[\hat{\beta}]-\beta_*\big\|_{\hat{\Sigma}}^2}_{\text{Bias}}+\underbrace{\mathbb{E}\big[\big\|\hat{\beta}-\mathbb{E}[\hat{\beta}]\big\|_{\hat{\Sigma}}^2\big]}_{\text{Variance}}.$$

The proof is,

$$\begin{aligned}\mathbb{E}\big[R(\hat\beta)\big]-R^* &= \mathbb{E}\big[\big\|\hat\beta-\mathbb{E}[\hat\beta]+\mathbb{E}[\hat\beta]-\beta_*\big\|_{\hat\Sigma}^2\big]\\ &= \mathbb{E}\big[\big\|\hat\beta-\mathbb{E}[\hat\beta]\big\|_{\hat\Sigma}^2\big]+2\mathbb{E}\big[\big(\hat\beta-\mathbb{E}[\hat\beta]\big)^\top\hat\Sigma(\mathbb{E}[\hat\beta]-\beta_*)\big]+\mathbb{E}\big[\big\|\mathbb{E}[\hat\beta]-\beta_*\big\|_{\hat\Sigma}^2\big]\\ &= \mathbb{E}\big[\big\|\hat\beta-\mathbb{E}[\hat\beta]\big\|_{\hat\Sigma}^2\big]+0+\big\|\mathbb{E}[\hat\beta]-\beta_*\big\|_{\hat\Sigma}^2.\end{aligned}$$

OLS estimator $\hat\beta$ out-of-sample expected excess risk:

$$\mathbb{E}\big[R(\hat\beta)\big]=\mathbb{E}_Y\big[R(\hat\beta)\big]=\mathbb{E}_Y\big[\mathbb{E}_{Y_{\text{new}}}\big[\frac{\|Y_{\text{new}}-X\hat\beta\|^2}{n}\big]\big]=\sigma^2\big(1+\frac{d}{n}\big)$$

$$\begin{aligned}&\mathbb{E}\big[R(\hat\beta)\big]-R(\beta^*)\\ &=\mathbb{E}_Y\big\|\hat\beta-\beta^*\big\|_{\hat\Sigma}^2\\ &=\mathbb{E}_Y\big\|\hat\beta-E[\hat\beta]+E[\hat\beta]-\beta^*\big\|_{\hat\Sigma}^2\\ &=\underbrace{\mathbb{E}_Y\big\|\hat\beta-E[\hat\beta]\big\|_{\hat\Sigma}^2}_{\text{variance}}+\underbrace{\mathbb{E}_Y\big\|E[\hat\beta]-\beta^*\big\|_{\hat\Sigma}^2}_{\text{bias}^2}, \qquad \text{(Bias-variance tradeoff)}\\ &=\mathbb{E}_Y\big\|\hat\beta-E[\hat\beta]\big\|_{\hat\Sigma}^2\\ &=\sigma^2\frac{d}{n}\end{aligned}$$

OLS estimator $\hat\beta$ in-sample expected excess risk:

$$\mathbb{E}\big[R(\hat\beta)\big]=\mathbb{E}_{Y_{tr}}\big[R(\hat\beta)\big]=\mathbb{E}_{Y_{tr}}\big[\mathbb{E}_{Y_{tr}}\big[\frac{\|Y_{tr}-X\hat\beta\|^2}{n}\big]\big]=\sigma^2\big(1-\frac{d}{n}\big)$$

$$\begin{aligned}&\mathbb{E}\big[R(\hat\beta)\big]\\ &=\mathbb{E}\big[\frac{\big\|Y-X(X^\top X)^{-1}X^\top Y\big\|^2}{n}\big]\\ &=\mathbb{E}\big[\frac{\big\|(I-P_X)Y\big\|^2}{n}\big]\\ &=\mathbb{E}\big[\frac{\big\|(I-P_X)(X\beta+\epsilon)\big\|^2}{n}\big]\\ &=\mathbb{E}\big[\frac{\epsilon^T(I-P_X)\epsilon}{n}\big]\\ &=\frac{\mathbb{E}\big[\text{trace}((I-P_X)\epsilon^T\epsilon)\big]}{n}\\ &=\frac{\text{trace}[(I-P_X)E[\epsilon^T\epsilon]]}{n}\\ &=\sigma^2\big(1-\frac{d}{n}\big)\end{aligned}$$

Random design decomposition

Risk in random design

Model setup. We still assume a well-specified linear regression model, but allow the covariates to be random. The model is:

$$Y_i = X_i^\top \beta^* + \varepsilon_i, \qquad i = 1, \dots, n,$$

where $X_i \sim$ i.i.d. P_X are random vectors in $\mathbb{R}^d$, and the errors satisfy

$$\varepsilon_1, \dots, \varepsilon_n \mid X_1, \dots, X_n \;\sim\; \text{i.i.d. } (0, \sigma^2).$$

We observe n i.i.d. pairs $(Y_1, X_1), \dots, (Y_n, X_n) \in \mathbb{R} \times \mathbb{R}^d$.
Risk function. The risk function is defined as:

$$\beta \in \mathbb{R}^d \;\longmapsto\; R(\beta) = \mathbb{E}\big[(Y - X^\top \beta)^2\big],$$

where the expectation is over a new i.i.d. draw $(Y_{\text{new}}, X_{\text{new}}) \sim P_{Y,X}$, independent of the training data. Equivalently, the expectation can be written over (ε, X) since $Y = X^\top \beta^* + \varepsilon$.
Proposition. (Expression for the predictive risk) Let $\Sigma = \mathbb{E}\big[XX^\top\big] \succeq 0$. Then for all $\beta \in \mathbb{R}^d$,

$$R(\beta) \;=\; \big\|\beta - \beta^*\big\|_\Sigma^2 + \sigma^2.$$

Proof. Write $Y - X^\top\beta = (Y - X^\top\beta^*) + X^\top(\beta^* - \beta) = \varepsilon + X^\top(\beta^* - \beta)$. Expanding the square:

$$R(\beta) = \mathbb{E}\big[(Y - X^\top\beta)^2\big] = \underbrace{\mathbb{E}\big[\varepsilon^2\big]}_{\sigma^2} + \underbrace{\mathbb{E}\big[(X^\top(\beta^* - \beta))^2\big]}_{\|\beta - \beta^*\|_\Sigma^2} + 2\,\underbrace{\mathbb{E}\big[\varepsilon \cdot X^\top(\beta^* - \beta)\big]}_{\text{CV}}.$$

Cross term CV $= 0$. By the law of iterated expectations and $\mathbb{E}\big[\varepsilon \mid X\big] = 0$:

$$\begin{aligned}
\mathbb{E}\big[\text{CV}\big] &= \mathbb{E}\big[\mathbb{E}\big[\text{CV} \mid X\big]\big] \\
&= \mathbb{E}_X\big[\mathbb{E}_{\varepsilon|X}\big[\varepsilon \cdot X^\top(\beta^* - \beta)\big]\big] \\
&= \mathbb{E}_X\big[X^\top(\beta^* - \beta) \cdot \underbrace{\mathbb{E}_{\varepsilon|X}\big[\varepsilon\big]}_{=0}\big] \;=\; 0.
\end{aligned}$$

Therefore $R(\beta) = \big\|\beta - \beta^*\big\|_\Sigma^2 + \sigma^2$.
<u>*Remark.*</u> If the model is misspecified, β^* is the projection parameter: the coefficients of the L_2 projection of Y (or of $\mathbb{E}\big[Y \mid X\big]$) onto the linear span of X, i.e.

$$\beta^* = \operatorname*{argmin}_{\beta \in \mathbb{R}^d} \mathbb{E}\big[(Y - X^\top\beta)^2\big] = \Sigma^{-1}\mathbb{E}\big[Y \cdot X\big] \quad (\text{assuming } \Sigma \succ 0,\ \mathbb{E}\big[Y^2\big] < \infty).$$

In this case, CV $= 0$ still holds, by the defining property of L_2 projection: $Y - X^\top\beta^*$ is **uncorrelated** with any linear function of X. More precisely, $Y - X^\top\beta^*$ decomposes as

$$Y - X^\top\beta^* = \underbrace{\big(Y - \mathbb{E}\big[Y \mid X\big]\big)}_{\varepsilon,\ \perp f(X)\ \forall f} + \underbrace{\big(\mathbb{E}\big[Y \mid X\big] - X^\top\beta^*\big)}_{\eta,\ \perp \text{ linear span of } X},$$

and $X^\top(\beta^* - \beta)$ belongs to the linear span of X, so both terms are orthogonal to it. Therefore CV $= \mathbb{E}\big[(Y - X^\top\beta^*)(X^\top(\beta^* - \beta))\big] = 0$. In this case:

$$R(\beta) = \big\|\beta - \beta^*\big\|_\Sigma^2 + \sigma^2 + \eta^2,$$

where $\sigma^2 = \mathbb{E}\big[(Y - \mathbb{E}\big[Y \mid X\big])^2\big]$ and $\eta^2 = \mathbb{E}\big[(\mathbb{E}\big[Y \mid X\big] - X^\top\beta^*)^2\big]$ is the non-linearity term.

Expected excess risk

Excess risk. The excess risk is

$$R(\beta) - R(\beta^*) = \left\|\beta - \beta^*\right\|_\Sigma^2 \;\geq\; 0.$$

All we can do to minimize the risk is to minimize $\left\|\beta - \beta^*\right\|_\Sigma^2$, since σ^2 (or $\sigma^2 + \eta^2$) does not depend on β. Given data $(Y_i, X_i)_{i=1}^n$, the OLS estimator is:

$$\hat\beta = \hat\Sigma^{-1}\frac{1}{n}\sum_{i=1}^n Y_i X_i, \qquad \hat\Sigma = \frac{1}{n}\sum_{i=1}^n X_i X_i^\top = \hat{\mathbb{E}}_n\big[XX^\top\big].$$

The excess risk of $\hat\beta$ is $R(\hat\beta) - R(\beta^*) = \left\|\hat\beta - \beta^*\right\|_\Sigma^2$, which is a random variable.

Expected excess risk of OLS $\hat\beta$ in random design. Assume $\hat\Sigma$ is invertible. Then the expected excess risk of the OLS estimator $\hat\beta$ is:

$$\mathbb{E}\big[R(\hat\beta)\big] - R(\beta^*) = \mathbb{E}\big[\left\|\hat\beta - \beta^*\right\|_\Sigma^2\big] = \frac{\sigma^2}{n}\,\mathbb{E}_X\big[\operatorname{tr}\big(\Sigma\hat\Sigma^{-1}\big)\big].$$

Proof. Write X for the $n\times d$ matrix with rows $X_1^\top, \dots, X_n^\top$, so $\hat\Sigma = \frac{1}{n}X^\top X$. Similarly let $Y = (Y_1, \dots, Y_n)^\top$ and $\varepsilon = (\varepsilon_1, \dots, \varepsilon_n)^\top \in \mathbb{R}^n$. Since $Y = X\beta^* + \varepsilon$:

$$\hat\beta - \beta^* = \hat\Sigma^{-1}\frac{X^\top Y}{n} - \beta^* = \hat\Sigma^{-1}\frac{X^\top(X\beta^* + \varepsilon)}{n} - \beta^* = \hat\Sigma^{-1}\frac{X^\top\varepsilon}{n}.$$

Therefore the expected excess risk is:

$$\begin{aligned}
\mathbb{E}\big[\left\|\hat\beta - \beta^*\right\|_\Sigma^2\big] &= \mathbb{E}\Big[\frac{\varepsilon^\top X}{n}\hat\Sigma^{-1}\Sigma\hat\Sigma^{-1}\frac{X^\top\varepsilon}{n}\Big] \\
&= \mathbb{E}\Big[\operatorname{tr}\Big(\frac{\varepsilon^\top X}{n}\hat\Sigma^{-1}\Sigma\hat\Sigma^{-1}\frac{X^\top\varepsilon}{n}\Big)\Big] \\
&= \mathbb{E}\Big[\operatorname{tr}\Big(\Sigma\hat\Sigma^{-1}\frac{X^\top\varepsilon\varepsilon^\top X}{n^2}\hat\Sigma^{-1}\Big)\Big],
\end{aligned}$$

where the second line uses scalar = tr(scalar) and the third uses the cyclic property of trace. Now, conditioning on X and using $\mathbb{E}\big[\varepsilon\varepsilon^\top \mid X\big] = \sigma^2 I_n$:

$$\begin{aligned}
&= \mathbb{E}_X\Big[\operatorname{tr}\Big(\Sigma\hat\Sigma^{-1}\frac{X^\top\overbrace{\mathbb{E}\big[\varepsilon\varepsilon^\top \mid X\big]}^{\sigma^2 I_n}X}{n^2}\hat\Sigma^{-1}\Big)\Big] \\
&= \frac{\sigma^2}{n}\,\mathbb{E}_X\Big[\operatorname{tr}\Big(\Sigma\hat\Sigma^{-1}\underbrace{\frac{X^\top X}{n}}_{\hat\Sigma}\hat\Sigma^{-1}\Big)\Big] \\
&= \frac{\sigma^2}{n}\,\mathbb{E}_X\big[\operatorname{tr}\big(\Sigma\hat\Sigma^{-1}\big)\big].
\end{aligned}$$

Remark. $\mathbb{E}\big[\operatorname{tr}(\Sigma\hat\Sigma^{-1})\big] \geq d$ because on the cone of PD matrices the map $A \mapsto \operatorname{tr}(A^{-1})$ is convex, so by Jensen's inequality:

$$\mathbb{E}\big[\operatorname{tr}(\Sigma\hat\Sigma^{-1})\big] \;\geq\; \operatorname{tr}\Big(\Sigma\left(\mathbb{E}[\hat\Sigma]\right)^{-1}\Big) = \operatorname{tr}(\Sigma\cdot\Sigma^{-1}) = d.$$

Ridge risk decomposition

Expected excess risk $\hat\beta_\lambda$ (fixed-design)

Under the linear model $Y = X\beta_* + \varepsilon$ with fixed design $X \in \mathbb{R}^{n\times d}$ and $\varepsilon \sim$ i.i.d.$(0, \sigma^2)$, the ridge estimator $\hat\beta_\lambda = (\hat\Sigma + \lambda I)^{-1}\frac{X^\top Y}{n}$ has the following expected excess risk:

$$\mathbb{E}\big[R(\hat\beta_\lambda)\big] - R^* = \lambda^2 \beta_*^\top (\hat\Sigma + \lambda I)^{-2}\hat\Sigma\,\beta_* + \frac{\sigma^2}{n}\operatorname{tr}\big[\hat\Sigma^2(\hat\Sigma + \lambda I)^{-2}\big].$$

Proof. We use the expected excess risk decomposition from Section 8 (bias-variance tradeoff for any estimator $\hat\beta$):

$$\mathbb{E}\big[R(\hat\beta)\big] - R^* = \underbrace{\big\|\mathbb{E}[\hat\beta] - \beta_*\big\|_{\hat\Sigma}^2}_{B} + \underbrace{\mathbb{E}\big[\|\hat\beta - \mathbb{E}[\hat\beta]\|_{\hat\Sigma}^2\big]}_{V}.$$

We compute each term for $\hat\beta_\lambda$.
Step 1: Compute $\mathbb{E}[\hat\beta_\lambda]$. Substituting $Y = X\beta_* + \varepsilon$ and using $\mathbb{E}[\varepsilon] = 0$:

$$\mathbb{E}\big[\hat\beta_\lambda\big] = (\hat\Sigma + \lambda I)^{-1}\frac{X^\top X\beta_*}{n} = (\hat\Sigma + \lambda I)^{-1}\hat\Sigma\,\beta_*.$$

Writing $\hat\Sigma = (\hat\Sigma + \lambda I) - \lambda I$:

$$(\hat\Sigma + \lambda I)^{-1}\hat\Sigma = I - \lambda(\hat\Sigma + \lambda I)^{-1} \quad\Rightarrow\quad \mathbb{E}\big[\hat\beta_\lambda\big] - \beta_* = -\lambda(\hat\Sigma + \lambda I)^{-1}\beta_*.$$

Step 2: Bias term B. Using the fact that $\hat\Sigma$ and $(\hat\Sigma + \lambda I)^{-1}$ commute (see Remarks below):

$$\begin{aligned} B = \big\|\mathbb{E}[\hat\beta_\lambda] - \beta_*\big\|_{\hat\Sigma}^2 &= \lambda^2\beta_*^\top(\hat\Sigma + \lambda I)^{-1}\hat\Sigma(\hat\Sigma + \lambda I)^{-1}\beta_* \\ &= \lambda^2\beta_*^\top(\hat\Sigma + \lambda I)^{-2}\hat\Sigma\,\beta_*. \end{aligned}$$

The last step uses $(\hat\Sigma + \lambda I)^{-1}\hat\Sigma(\hat\Sigma + \lambda I)^{-1} = \hat\Sigma(\hat\Sigma + \lambda I)^{-2}$ by commutativity.
Step 3: Variance term V. Since $\hat\beta_\lambda - \mathbb{E}[\hat\beta_\lambda] = (\hat\Sigma + \lambda I)^{-1}\frac{X^\top\varepsilon}{n}$, we expand the $\hat\Sigma$-norm as a trace and apply the cyclic property:

$$\begin{aligned} V &= \mathbb{E}\Big[\Big\|\tfrac{1}{n}(\hat\Sigma + \lambda I)^{-1}X^\top\varepsilon\Big\|_{\hat\Sigma}^2\Big] \\ &= \mathbb{E}\Big[\frac{1}{n^2}\operatorname{tr}\Big(\varepsilon^\top X(\hat\Sigma + \lambda I)^{-1}\hat\Sigma(\hat\Sigma + \lambda I)^{-1}X^\top\varepsilon\Big)\Big] \\ &= \mathbb{E}\Big[\frac{1}{n^2}\operatorname{tr}\Big(X^\top\varepsilon\varepsilon^\top X(\hat\Sigma + \lambda I)^{-1}\hat\Sigma(\hat\Sigma + \lambda I)^{-1}\Big)\Big] \\ &= \frac{1}{n^2}\operatorname{tr}\Big(X^\top\underbrace{\mathbb{E}[\varepsilon\varepsilon^\top]}_{\sigma^2 I}X(\hat\Sigma + \lambda I)^{-1}\hat\Sigma(\hat\Sigma + \lambda I)^{-1}\Big) \\ &= \frac{\sigma^2}{n}\operatorname{tr}\Big(\hat\Sigma(\hat\Sigma + \lambda I)^{-1}\hat\Sigma(\hat\Sigma + \lambda I)^{-1}\Big) \\ &= \frac{\sigma^2}{n}\operatorname{tr}\big[\hat\Sigma^2(\hat\Sigma + \lambda I)^{-2}\big]. \end{aligned}$$

The last step again uses commutativity to merge the two factors.
Conclusion. Summing the bias and variance terms:

$$\mathbb{E}\big[R(\hat\beta_\lambda)\big] - R^* = \lambda^2 \beta_*^\top (\hat\Sigma + \lambda I)^{-2}\hat\Sigma\,\beta_* + \frac{\sigma^2}{n}\operatorname{tr}\big[\hat\Sigma^2(\hat\Sigma + \lambda I)^{-2}\big].$$

Remarks.

Remark 1. (Commutativity of $\hat{\Sigma}$ and $(\hat{\Sigma}+\lambda I)^{-1}$ algebraic proof.) We want to show $\hat{\Sigma}(\hat{\Sigma}+\lambda I)^{-1} = (\hat{\Sigma}+\lambda I)^{-1}\hat{\Sigma}$. This is equivalent to showing $(\hat{\Sigma}+\lambda I)\hat{\Sigma} = \hat{\Sigma}(\hat{\Sigma}+\lambda I)$. Expanding both sides:

$$(\hat{\Sigma}+\lambda I)\hat{\Sigma} = \hat{\Sigma}^2 + \lambda\hat{\Sigma}, \qquad \hat{\Sigma}(\hat{\Sigma}+\lambda I) = \hat{\Sigma}^2 + \lambda\hat{\Sigma}.$$

Both sides are equal, so $(\hat{\Sigma}+\lambda I)$ and $\hat{\Sigma}$ commute. For invertible matrices, $AB = BA$ implies $A^{-1}B = BA^{-1}$ (left-multiply by A^{-1} and right-multiply by A^{-1}), so $(\hat{\Sigma}+\lambda I)^{-1}$ and $\hat{\Sigma}$ also commute.

Remark 2. (Commutativity of $\hat{\Sigma}$ and $(\hat{\Sigma}+\lambda I)^{-1}$ — eigendecomposition perspective.) Since $\hat{\Sigma}$ is symmetric PSD, it admits the eigendecomposition $\hat{\Sigma} = Q\Lambda Q^\top$, where Q is orthogonal and $\Lambda = \mathrm{diag}(\mu_1, \ldots, \mu_d)$. Then:

$$(\hat{\Sigma}+\lambda I)^{-1} = \left(Q\Lambda Q^\top + \lambda I\right)^{-1} = Q(\Lambda+\lambda I)^{-1}Q^\top.$$

Both matrices share the same eigenvectors Q:

$$\hat{\Sigma} = Q\Lambda Q^\top, \qquad (\hat{\Sigma}+\lambda I)^{-1} = Q(\Lambda+\lambda I)^{-1}Q^\top.$$

So when multiplying:

$$\hat{\Sigma}\cdot(\hat{\Sigma}+\lambda I)^{-1} = Q\Lambda Q^\top \cdot Q(\Lambda+\lambda I)^{-1}Q^\top = Q \underbrace{\Lambda(\Lambda+\lambda I)^{-1}}_{\text{diagonal, commutes}} Q^\top,$$

$$(\hat{\Sigma}+\lambda I)^{-1}\cdot\hat{\Sigma} = Q(\Lambda+\lambda I)^{-1}Q^\top \cdot Q\Lambda Q^\top = Q \underbrace{(\Lambda+\lambda I)^{-1}\Lambda}_{\text{same as above}} Q^\top.$$

Since Λ and $(\Lambda+\lambda I)^{-1}$ are both diagonal matrices, they commute trivially. Therefore $\hat{\Sigma}$ and $(\hat{\Sigma}+\lambda I)^{-1}$ commute.

More generally, $f(\hat{\Sigma})$ and $g(\hat{\Sigma})$ commute for any matrix functions f and g, since both are diagonalized by the same Q.

Expected excess risk $\hat{\beta}_\lambda$ (random-design) Method 1

Under the same random-design assumptions as Method 2, the ridge estimator $\hat{\beta}_\lambda = (\hat{\Sigma} + \lambda I)^{-1} \frac{X^\top Y}{n}$ has the following expected excess risk:

$$\mathbb{E}\big[R(\hat{\beta}_\lambda)\big] - R^* = \lambda^2\, \mathbb{E}\big[\beta_*^\top (\hat{\Sigma} + \lambda I)^{-1} \Sigma (\hat{\Sigma} + \lambda I)^{-1} \beta_*\big] + \frac{\sigma^2}{n}\, \mathbb{E}\big[\mathrm{tr}\big[(\hat{\Sigma} + \lambda I)^{-2} \hat{\Sigma}\Sigma\big]\big].$$

Proof. Recall $R(\beta) - R^* = \big\|\beta - \beta_*\big\|_\Sigma^2$ in random design. Thus

$$\mathbb{E}\big[R(\hat{\beta}_\lambda)\big] - R^* = \mathbb{E}\big[\big\|\hat{\beta}_\lambda - \beta_*\big\|_\Sigma^2\big].$$

Step 1: Decompose $\hat{\beta}_\lambda - \beta_*$. Substituting $Y = X\beta_* + \varepsilon$:

$$\hat{\beta}_\lambda = (\hat{\Sigma} + \lambda I)^{-1} \frac{X^\top X \beta_* + X^\top \varepsilon}{n} = (\hat{\Sigma} + \lambda I)^{-1} \hat{\Sigma}\, \beta_* + (\hat{\Sigma} + \lambda I)^{-1} \frac{X^\top \varepsilon}{n}.$$

Using $(\hat{\Sigma} + \lambda I)^{-1} \hat{\Sigma} = I - \lambda (\hat{\Sigma} + \lambda I)^{-1}$:

$$\hat{\beta}_\lambda - \beta_* = \underbrace{-\lambda (\hat{\Sigma} + \lambda I)^{-1} \beta_*}_{=: a(X)} + \underbrace{(\hat{\Sigma} + \lambda I)^{-1} \frac{X^\top \varepsilon}{n}}_{=: b(X, \varepsilon)}.$$

Step 2: Expand $\big\|\cdot\big\|_\Sigma^2$ and take expectation.

$$\big\|\hat{\beta}_\lambda - \beta_*\big\|_\Sigma^2 = \big\|a\big\|_\Sigma^2 + \big\|b\big\|_\Sigma^2 + 2\, a^\top \Sigma\, b.$$

Taking expectation:

$$\mathbb{E}\big[\big\|\hat{\beta}_\lambda - \beta_*\big\|_\Sigma^2\big] = \underbrace{\mathbb{E}\big[\big\|a\big\|_\Sigma^2\big]}_{B} + \underbrace{\mathbb{E}\big[\big\|b\big\|_\Sigma^2\big]}_{V} + 2\, \mathbb{E}\big[a^\top \Sigma\, b\big].$$

Step 3: Cross term vanishes via tower property. Apply tower property $\mathbb{E}[\cdot] = \mathbb{E}_X\big[\mathbb{E}[\cdot \mid X]\big]$. Given X, both a and Σ are deterministic, so they pull out of $\mathbb{E}[\cdot \mid X]$:

$$\mathbb{E}\big[a^\top \Sigma\, b\big] = \mathbb{E}_X\big[a^\top \Sigma\, \mathbb{E}[b \mid X]\big] = \mathbb{E}_X\Big[a^\top \Sigma\, (\hat{\Sigma} + \lambda I)^{-1} \frac{X^\top}{n} \underbrace{\mathbb{E}[\varepsilon \mid X]}_{=0}\Big] = 0.$$

Step 4: Bias term B. Direct substitution of $a = -\lambda (\hat{\Sigma} + \lambda I)^{-1} \beta_*$:

$$B = \mathbb{E}\big[\big\|a\big\|_\Sigma^2\big] = \lambda^2\, \mathbb{E}\big[\beta_*^\top (\hat{\Sigma} + \lambda I)^{-1} \Sigma (\hat{\Sigma} + \lambda I)^{-1} \beta_*\big].$$

Step 5: Variance term V. Expand the Σ-norm as a trace, apply cyclic property, and use tower property with $\mathbb{E}\big[\varepsilon\varepsilon^\top \mid X\big] = \sigma^2 I_n$:

$$\begin{aligned}
V &= \frac{1}{n^2}\, \mathbb{E}\Big[\mathrm{tr}\Big(X^\top \varepsilon\varepsilon^\top X\, (\hat{\Sigma} + \lambda I)^{-1} \Sigma (\hat{\Sigma} + \lambda I)^{-1}\Big)\Big] \\
&= \frac{1}{n^2}\, \mathbb{E}_X\Big[\mathrm{tr}\Big(X^\top \underbrace{\mathbb{E}\big[\varepsilon\varepsilon^\top \mid X\big]}_{\sigma^2 I_n} X\, (\hat{\Sigma} + \lambda I)^{-1} \Sigma (\hat{\Sigma} + \lambda I)^{-1}\Big)\Big] \\
&= \frac{\sigma^2}{n}\, \mathbb{E}\big[\mathrm{tr}\big(\hat{\Sigma}\, (\hat{\Sigma} + \lambda I)^{-1} \Sigma (\hat{\Sigma} + \lambda I)^{-1}\big)\big].
\end{aligned}$$

Using cyclic property of trace and commutativity of $\hat{\Sigma}$ with $(\hat{\Sigma} + \lambda I)^{-1}$:

$$\mathrm{tr}\big[\hat{\Sigma} (\hat{\Sigma} + \lambda I)^{-1} \Sigma (\hat{\Sigma} + \lambda I)^{-1}\big] = \mathrm{tr}\big[(\hat{\Sigma} + \lambda I)^{-1} \hat{\Sigma} (\hat{\Sigma} + \lambda I)^{-1} \Sigma\big] = \mathrm{tr}\big[(\hat{\Sigma} + \lambda I)^{-2} \hat{\Sigma}\, \Sigma\big].$$

Therefore:

$$V = \frac{\sigma^2}{n}\,\mathbb{E}\big[\operatorname{tr}\big[(\hat{\Sigma} + \lambda I)^{-2}\hat{\Sigma}\Sigma\big]\big].$$

Conclusion.

$$\mathbb{E}\big[R(\hat{\beta}_\lambda)\big] - R^* = \lambda^2\,\mathbb{E}\big[\beta_*^\top(\hat{\Sigma} + \lambda I)^{-1}\Sigma(\hat{\Sigma} + \lambda I)^{-1}\beta_*\big] + \frac{\sigma^2}{n}\,\mathbb{E}\big[\operatorname{tr}\big[(\hat{\Sigma} + \lambda I)^{-2}\hat{\Sigma}\Sigma\big]\big].$$

Expected excess risk $\hat{\beta}_\lambda$ (random-design) Method 2

Under the random-design linear model with i.i.d. pairs (Y_i, X_i), $i = 1, \dots, n$, satisfying $Y_i = X_i^\top \beta_* + \varepsilon_i$, $\mathbb{E}[\varepsilon_i \mid X_i] = 0$, $\mathrm{Var}(\varepsilon_i \mid X_i) = \sigma^2$, and $\Sigma = \mathbb{E}[XX^\top] \succ 0$, the ridge estimator $\hat{\beta}_\lambda = (\hat{\Sigma} + \lambda I)^{-1} \frac{X^\top Y}{n}$ has the following expected excess risk:

$$\mathbb{E}[R(\hat{\beta}_\lambda)] - R^* = \lambda^2\, \mathbb{E}[\beta_*^\top (\hat{\Sigma} + \lambda I)^{-1} \Sigma (\hat{\Sigma} + \lambda I)^{-1} \beta_*] + \frac{\sigma^2}{n}\, \mathbb{E}[\mathrm{tr}[(\hat{\Sigma} + \lambda I)^{-2} \hat{\Sigma} \Sigma]].$$

Proof. Recall $R(\beta) - R^* = \|\beta - \beta_*\|_\Sigma^2$ in random design (uses the *population* Σ, since $X_{\text{new}} \sim P_X$ is a fresh draw). Thus

$$\mathbb{E}[R(\hat{\beta}_\lambda)] - R^* = \mathbb{E}[\|\hat{\beta}_\lambda - \beta_*\|_\Sigma^2].$$

Since $\hat{\Sigma}$ is random, we condition on $X = (X_1, \dots, X_n)$ first and apply the bias-variance decomposition, then take expectation over X:

$$\mathbb{E}[\|\hat{\beta}_\lambda - \beta_*\|_\Sigma^2] = \underbrace{\mathbb{E}_X[\|\mathbb{E}[\hat{\beta}_\lambda \mid X] - \beta_*\|_\Sigma^2]}_{B} + \underbrace{\mathbb{E}[\|\hat{\beta}_\lambda - \mathbb{E}[\hat{\beta}_\lambda \mid X]\|_\Sigma^2]}_{V}.$$

The cross term vanishes because $\mathbb{E}[\hat{\beta}_\lambda \mid X] - \beta_*$ is deterministic given X, while $\hat{\beta}_\lambda - \mathbb{E}[\hat{\beta}_\lambda \mid X]$ has mean zero given X.

Step 1: Compute $\mathbb{E}[\hat{\beta}_\lambda \mid X]$. Substituting $Y = X\beta_* + \varepsilon$ and using $\mathbb{E}[\varepsilon \mid X] = 0$:

$$\mathbb{E}[\hat{\beta}_\lambda \mid X] = (\hat{\Sigma} + \lambda I)^{-1} \frac{X^\top X \beta_*}{n} = (\hat{\Sigma} + \lambda I)^{-1} \hat{\Sigma}\, \beta_*.$$

Writing $\hat{\Sigma} = (\hat{\Sigma} + \lambda I) - \lambda I$:

$$(\hat{\Sigma} + \lambda I)^{-1} \hat{\Sigma} = I - \lambda (\hat{\Sigma} + \lambda I)^{-1} \quad \Rightarrow \quad \mathbb{E}[\hat{\beta}_\lambda \mid X] - \beta_* = -\lambda (\hat{\Sigma} + \lambda I)^{-1} \beta_*.$$

Step 2: Bias term B.

$$\begin{aligned} B &= \mathbb{E}_X[\|\mathbb{E}[\hat{\beta}_\lambda \mid X] - \beta_*\|_\Sigma^2] \\ &= \mathbb{E}_X[\lambda^2 \beta_*^\top (\hat{\Sigma} + \lambda I)^{-1} \Sigma (\hat{\Sigma} + \lambda I)^{-1} \beta_*] \\ &= \lambda^2\, \mathbb{E}[\beta_*^\top (\hat{\Sigma} + \lambda I)^{-1} \Sigma (\hat{\Sigma} + \lambda I)^{-1} \beta_*]. \end{aligned}$$

Unlike the fixed-design case, we *cannot* merge $(\hat{\Sigma} + \lambda I)^{-1} \Sigma (\hat{\Sigma} + \lambda I)^{-1}$ into $\Sigma (\hat{\Sigma} + \lambda I)^{-2}$ because Σ and $\hat{\Sigma}$ do not commute in general (they are different matrices, not functions of the same matrix).

Step 3: Variance term V. Since $\hat{\beta}_\lambda - \mathbb{E}[\hat{\beta}_\lambda \mid X] = (\hat{\Sigma} + \lambda I)^{-1} \frac{X^\top \varepsilon}{n}$, we expand the Σ-norm as a trace, condition on X, and apply the cyclic property:

$$\begin{aligned} V &= \mathbb{E}\Big[\Big\|\tfrac{1}{n} (\hat{\Sigma} + \lambda I)^{-1} X^\top \varepsilon\Big\|_\Sigma^2\Big] \\ &= \mathbb{E}\Big[\frac{1}{n^2} \mathrm{tr}\Big(\varepsilon^\top X (\hat{\Sigma} + \lambda I)^{-1} \Sigma (\hat{\Sigma} + \lambda I)^{-1} X^\top \varepsilon\Big)\Big] \\ &= \mathbb{E}_X\Big[\frac{1}{n^2} \mathrm{tr}\Big(X^\top \underbrace{\mathbb{E}[\varepsilon\varepsilon^\top \mid X]}_{\sigma^2 I_n} X (\hat{\Sigma} + \lambda I)^{-1} \Sigma (\hat{\Sigma} + \lambda I)^{-1}\Big)\Big] \\ &= \frac{\sigma^2}{n}\, \mathbb{E}\Big[\mathrm{tr}\Big(\hat{\Sigma} (\hat{\Sigma} + \lambda I)^{-1} \Sigma (\hat{\Sigma} + \lambda I)^{-1}\Big)\Big]. \end{aligned}$$

Now using cyclic property of trace and commutativity of $\hat{\Sigma}$ with $(\hat{\Sigma} + \lambda I)^{-1}$ (both are functions of the same matrix):

$$\begin{aligned} \mathrm{tr}[\hat{\Sigma} (\hat{\Sigma} + \lambda I)^{-1} \Sigma (\hat{\Sigma} + \lambda I)^{-1}] &= \mathrm{tr}[(\hat{\Sigma} + \lambda I)^{-1} \hat{\Sigma} (\hat{\Sigma} + \lambda I)^{-1} \Sigma] \quad \text{(cyclic)} \\ &= \mathrm{tr}[(\hat{\Sigma} + \lambda I)^{-2} \hat{\Sigma}\, \Sigma] \quad \text{(commutativity)}. \end{aligned}$$

Therefore:
$$V = \frac{\sigma^2}{n}\,\mathbb{E}\big[\mathrm{tr}\big[(\hat\Sigma + \lambda I)^{-2}\hat\Sigma\Sigma\big]\big].$$

Conclusion. Summing the bias and variance terms:

$$\mathbb{E}\big[R(\hat\beta_\lambda)\big] - R^* = \lambda^2\,\mathbb{E}\big[\beta_*^\top(\hat\Sigma + \lambda I)^{-1}\Sigma(\hat\Sigma + \lambda I)^{-1}\beta_*\big] + \frac{\sigma^2}{n}\,\mathbb{E}\big[\mathrm{tr}\big[(\hat\Sigma + \lambda I)^{-2}\hat\Sigma\Sigma\big]\big].$$

Remarks.

Remark 1. (Comparison with fixed-design.) The random-design formula reduces to the fixed-design formula upon two substitutions:

- Replace the population Σ by the sample $\hat\Sigma$ (in fixed design, the risk is in-sample, so the norm is $\|\cdot\|_{\hat\Sigma}$).
- Drop the outer expectation $\mathbb{E}_X[\cdot]$ (in fixed design, X is deterministic, so $\hat\Sigma$ is non-random).

Under these substitutions, Σ becomes $\hat\Sigma$, which then commutes with $(\hat\Sigma + \lambda I)^{-1}$, and the bias term simplifies to $\lambda^2\beta_*^\top(\hat\Sigma + \lambda I)^{-2}\hat\Sigma\beta_*$, recovering the fixed-design expression.

Remark 2. (Why the bias term cannot be simplified.) In fixed-design, the bias term simplifies because $\hat\Sigma$ commutes with $(\hat\Sigma + \lambda I)^{-1}$ via the eigendecomposition perspective: both share the same eigenvectors. In random-design, the bias term involves Σ sandwiched between two $(\hat\Sigma + \lambda I)^{-1}$ factors. Since Σ and $\hat\Sigma$ have generally different eigenvectors, no such simplification is possible. The variance term, in contrast, only involves $\hat\Sigma$ and $(\hat\Sigma + \lambda I)^{-1}$ alongside Σ, and the commutativity between $\hat\Sigma$ and $(\hat\Sigma + \lambda I)^{-1}$ allows us to merge them via the cyclic trace property.

Remark 3. (Why condition on X — computational, not orthogonality.) In the bias-variance decomposition, we add and subtract $\mathbb{E}\big[\hat\beta_\lambda \mid X\big]$ rather than the unconditional $\mathbb{E}\big[\hat\beta_\lambda\big]$. This is *not* because the cross term would fail to vanish otherwise — adding and subtracting $\mathbb{E}\big[\hat\beta_\lambda\big]$ also gives a zero cross term, since $\mathbb{E}\big[\hat\beta_\lambda\big] - \beta_*$ is a constant and $\mathbb{E}\big[\hat\beta_\lambda - \mathbb{E}\big[\hat\beta_\lambda\big]\big] = 0$. The reason is **computational**: the conditional expectation has a closed-form expression, while the unconditional one does not:

$$\mathbb{E}\big[\hat\beta_\lambda \mid X\big] = (\hat\Sigma + \lambda I)^{-1}\hat\Sigma\,\beta_* \qquad \text{vs.} \qquad \mathbb{E}\big[\hat\beta_\lambda\big] = \mathbb{E}\big[(\hat\Sigma + \lambda I)^{-1}\hat\Sigma\big]\,\beta_* = \,?$$

The unconditional expectation involves $\mathbb{E}\big[(\hat\Sigma + \lambda I)^{-1}\hat\Sigma\big]$, i.e. the expectation of a nonlinear function of the random matrix $\hat\Sigma$, which has no tractable closed form. By conditioning on X, $\hat\Sigma$ becomes deterministic, and the bias $\mathbb{E}\big[\hat\beta_\lambda \mid X\big] - \beta_* = -\lambda(\hat\Sigma + \lambda I)^{-1}\beta_*$ is an explicit expression that we can directly substitute into subsequent steps.

Upper bound for ridge estimator β_λ expected risk

Choice of regularization parameter With the choice of the regularization parameter

$$\lambda_* = \frac{\sigma\,\mathrm{tr}(\hat\Sigma)^{1/2}}{\|\beta_*\|_2\sqrt{n}},$$

we have

$$\mathbb{E}\big[R(\hat\beta_{\lambda_*})\big] - R^* \;\le\; \frac{\sigma\,\mathrm{tr}(\hat\Sigma)^{1/2}\|\beta_*\|_2}{\sqrt{n}}.$$

Proof. From the expected risk decomposition, the excess risk decomposes as $B+V$ where

$$B = \lambda^2\beta_*^\top(\hat\Sigma+\lambda I)^{-2}\hat\Sigma\,\beta_*, \qquad V = \frac{\sigma^2}{n}\,\mathrm{tr}\big[\hat\Sigma^2(\hat\Sigma+\lambda I)^{-2}\big].$$

We bound each term separately.
Step 1: Key scalar inequality. For any $\mu \ge 0$ and $\lambda > 0$:

$$\frac{\mu\lambda}{(\mu+\lambda)^2} \le \frac{1}{2} \quad\Longleftrightarrow\quad (\mu+\lambda)^2 \ge 2\mu\lambda \quad\Longleftrightarrow\quad (\mu-\lambda)^2 \ge 0,$$

which is trivially true. This scalar fact will be applied to the eigenvalues of $\hat\Sigma$.
Step 2: Scalar inequality $\to$ matrix inequality. Let $\hat\Sigma = Q\Lambda Q^\top$ be the eigendecomposition with eigenvalues $\mu_1,\dots,\mu_d \ge 0$. Then:

$$(\hat\Sigma+\lambda I)^{-2}\lambda\hat\Sigma = Q\,\mathrm{diag}\big(\frac{\lambda\mu_i}{(\mu_i+\lambda)^2}\big)\,Q^\top \;\preceq\; \frac{1}{2}I,$$

since each diagonal entry satisfies $\frac{\lambda\mu_i}{(\mu_i+\lambda)^2} \le \frac{1}{2}$ by Step 1.
Step 3: Upper bound on B. Denote $M = (\hat\Sigma+\lambda I)^{-2}\lambda\hat\Sigma \preceq \frac{1}{2}I$. For any vector β_*, the PSD order gives $\beta_*^\top M\beta_* \le \frac{1}{2}\|\beta_*\|_2^2$, so:

$$B = \lambda\,\beta_*^\top\underbrace{(\hat\Sigma+\lambda I)^{-2}\lambda\hat\Sigma}_{M\preceq\frac{1}{2}I}\beta_* \;\le\; \frac{\lambda}{2}\|\beta_*\|_2^2.$$

Step 4: Upper bound on V. Extracting a factor of $\frac{1}{\lambda}$ and applying Step 2:

$$\hat\Sigma^2(\hat\Sigma+\lambda I)^{-2} = \frac{1}{\lambda}\hat\Sigma\cdot\underbrace{\lambda\hat\Sigma(\hat\Sigma+\lambda I)^{-2}}_{\preceq\frac{1}{2}I} \;\preceq\; \frac{\hat\Sigma}{2\lambda}.$$

Since trace is monotone with respect to the PSD order:

$$V = \frac{\sigma^2}{n}\,\mathrm{tr}\big[\hat\Sigma^2(\hat\Sigma+\lambda I)^{-2}\big] \;\le\; \frac{\sigma^2}{n}\cdot\frac{\mathrm{tr}(\hat\Sigma)}{2\lambda} = \frac{\sigma^2\,\mathrm{tr}(\hat\Sigma)}{2\lambda n}.$$

Step 5: Combine and optimize over λ. Adding the two bounds:

$$\mathbb{E}\big[R(\hat\beta_\lambda)\big] - R^* \;\le\; \underbrace{\frac{\|\beta_*\|_2^2}{2}}_{a}\,\lambda \;+\; \underbrace{\frac{\sigma^2\,\mathrm{tr}(\hat\Sigma)}{2n}}_{b}\cdot\frac{1}{\lambda}.$$

This is of the form $a\lambda + b/\lambda$ for $a, b > 0$. Setting $\frac{d}{d\lambda}(a\lambda+b/\lambda) = 0$ gives $\lambda_* = \sqrt{b/a}$, and the minimum value is $2\sqrt{ab}$:

$$\lambda_* = \sqrt{\frac{b}{a}} = \frac{\sigma\,\mathrm{tr}(\hat\Sigma)^{1/2}}{\|\beta_*\|_2\sqrt{n}}, \qquad \mathbb{E}\big[R(\hat\beta_{\lambda_*})\big] - R^* \;\le\; 2\sqrt{ab} = \frac{\sigma\,\mathrm{tr}(\hat\Sigma)^{1/2}\|\beta_*\|_2}{\sqrt{n}}.$$

9 OLS asymptotic normality in fixed design

Consistency of $\hat{\beta}$ under fixed design

Setup. Assume the linear model $Y = \Phi\beta^* + \varepsilon$ with fixed design matrix $\Phi \in \mathbb{R}^{n\times d}$, where ε_i are independent with $\mathbb{E}[\varepsilon_i] = 0$ and $\mathrm{Var}(\varepsilon_i) = \sigma^2$. Let $\hat{\Sigma} = \frac{\Phi^\top \Phi}{n} \to \Sigma$ (positive definite). We claim $\hat{\beta} \xrightarrow{p} \beta^*$.
Key tool: L^2 convergence implies convergence in probability. It suffices to show $\mathbb{E}\big[\|\hat{\beta} - \beta^*\|^2\big] \to 0$, since by Markov's inequality,

$$\mathbb{P}\{\|\hat{\beta} - \beta^*\| > \epsilon\} \le \frac{\mathbb{E}\big[\|\hat{\beta} - \beta^*\|^2\big]}{\epsilon^2} \to 0.$$

To verify L^2 convergence, we use the MSE decomposition:

$$\mathbb{E}\big[\|\hat{\beta} - \beta^*\|^2\big] = \mathrm{Bias}^2(\hat{\beta}) + \mathrm{Var}(\hat{\beta}) \to 0.$$

Note that unbiasedness is not necessary; it suffices that $\mathrm{Bias}(\hat{\beta}) \to 0$ and $\mathrm{Var}(\hat{\beta}) \to 0$.
Expanding $\hat{\beta} - \beta^*$. Substituting $Y = \Phi\beta^* + \varepsilon$:

$$\hat{\beta} - \beta^* = \left(\Phi^\top \Phi\right)^{-1} \Phi^\top \varepsilon = \hat{\Sigma}^{-1} \frac{\Phi^\top \varepsilon}{n}.$$

Hence $\hat{\beta} \xrightarrow{p} \beta^*$ if and only if $\hat{\Sigma}^{-1}\frac{\Phi^\top \varepsilon}{n} \xrightarrow{p} 0$.

Method 1 (Direct). Since Φ_i is fixed, $\mathbb{E}\big[\hat{\beta} - \beta^*\big] = \hat{\Sigma}^{-1}\frac{\Phi^\top \mathbb{E}[\varepsilon]}{n} = 0$, so $\hat{\beta}$ is unbiased and Bias $= 0$. The variance is:

$$\mathrm{Var}\big(\hat{\beta}\big) = \sigma^2 \left(\Phi^\top \Phi\right)^{-1} = \frac{\sigma^2}{n}\hat{\Sigma}^{-1} \to 0.$$

By L^2 convergence, $\hat{\beta} \xrightarrow{p} \beta^*$.
Method 2 (Modular, via Slutsky).
Module 1: Show $\frac{\Phi^\top \varepsilon}{n} \xrightarrow{p} 0$. We have $\mathbb{E}\big[\frac{\Phi^\top \varepsilon}{n}\big] = 0$ and

$$\mathrm{Var}\Big[\frac{\Phi^\top \varepsilon}{n}\Big] = \frac{\sigma^2}{n^2}\Phi^\top \Phi = \frac{\sigma^2}{n}\hat{\Sigma} = \frac{\sigma^2}{n}\Sigma + \frac{\sigma^2}{n}\underbrace{\big(\hat{\Sigma} - \Sigma\big)}_{\to 0} \to 0.$$

By L^2 convergence, $\frac{\Phi^\top \varepsilon}{n} \xrightarrow{p} 0$.
Module 2: $\hat{\Sigma}^{-1}$ is bounded. Since Σ is positive definite, Σ is invertible, so $\hat{\Sigma}^{-1} \xrightarrow{p} \Sigma^{-1}$ and $\|\hat{\Sigma}^{-1}\|$ does not blow up as $n \to \infty$.
Slutsky's theorem:

$$\hat{\Sigma}^{-1} \cdot \frac{\Phi^\top \varepsilon}{n} \xrightarrow{p} \Sigma^{-1} \cdot 0 = 0, \quad \text{hence} \quad \hat{\beta} \xrightarrow{p} \beta^*.$$

<u>**Remark.**</u> Method 2's modular structure directly extends to the CLT proof: replacing $\xrightarrow{p} 0$ with $\xrightarrow{d} \mathcal{N}_d(0, \sigma^2\Sigma)$ yields asymptotic normality of $\hat{\beta}$.

Asymptotic Normality for $\hat{\beta}$ under fixed design setting

Setup. Assume the linear model $Y = \Phi\beta^* + \varepsilon$ with fixed design matrix $\Phi \in \mathbb{R}^{n\times d}$, where ε_i are independent with $\mathbb{E}[\varepsilon_i] = 0$ and $\mathrm{Var}(\varepsilon_i) = \sigma^2$. Let $\hat{\Sigma} = \frac{\Phi^\top \Phi}{n}$. We aim to show:

$$\sqrt{n}(\hat{\beta} - \beta^*) \xrightarrow{d} \mathcal{N}_d(0, \sigma^2\Sigma^{-1}),$$

under the conditions: (i) $\hat{\Sigma} \to \Sigma$ (positive definite), and (ii) $\max_i \frac{\|\Phi_i\|}{\sqrt{n}} \to 0$ (UAN condition).

Step 1: Expand $\hat\beta - \beta^*$. Since $\hat\beta = \hat\Sigma^{-1}\frac{\Phi^\top Y}{n}$, substituting $Y = \Phi\beta^* + \varepsilon$ and using $\hat\Sigma^{-1}\hat\Sigma = I_d$:

$$\sqrt{n}(\hat\beta - \beta^*) = \hat\Sigma^{-1} \cdot \sqrt{n}\,\frac{\Phi^\top \varepsilon}{n}.$$

Step 2: Reduction via Slutsky. Since $\hat\Sigma^{-1} \xrightarrow{p} \Sigma^{-1}$, by Slutsky's theorem it suffices to show:

$$\sqrt{n}\,\frac{\Phi^\top \varepsilon}{n} \xrightarrow{d} \mathcal{N}_d(0,\, \sigma^2\Sigma).$$

Step 3: Cast as a triangular array. Let $X_{ni} = \Phi_i\varepsilon_i \in \mathbb{R}^d$, where Φ_i is the i-th row of Φ (as a column vector). Define $z_n = \frac{1}{n}\sum_{i=1}^n X_{ni} = \frac{1}{n}\sum_{i=1}^n \Phi_i\epsilon_i$. Since Φ_i is fixed:

- $\mathbb{E}\big[X_{ni}\big] = \Phi_i\mathbb{E}\big[\varepsilon_i\big] = 0$.
- $\mathrm{Var}(X_{ni}) = \sigma^2\Phi_i\Phi_i^\top$.
- $V_n = \frac{1}{n}\sum_{i=1}^n \mathrm{Var}(X_{ni}) = \sigma^2\frac{\Phi^\top\Phi}{n} \to \sigma^2\Sigma =: V$.

Step 4: Verify the Lindeberg–Feller condition. We need to show: for every $\eta > 0$,

$$\frac{1}{n}\sum_{i=1}^n \mathbb{E}\big[\big\|\Phi_i\varepsilon_i\big\|^2 \cdot \mathbf{1}\big\{\big\|\Phi_i\varepsilon_i\big\| \ge \eta\sqrt{n}\big\}\big] \to 0.$$

Since Φ_i is fixed and $\big\|\Phi_i\varepsilon_i\big\|^2 = \big\|\Phi_i\big\|^2\varepsilon_i^2$, we extract $\big\|\Phi_i\big\|^2$ from the expectation. The event $\big\{\big\|\Phi_i\varepsilon_i\big\| \ge \eta\sqrt{n}\big\}$ is equivalent to $\big\{\big|\varepsilon_i\big| \ge \frac{\eta\sqrt{n}}{\|\Phi_i\|}\big\}$. Bounding by the max over i:

$$\le \underbrace{\frac{1}{n}\sum_{i=1}^n \big\|\Phi_i\big\|^2}_{=\frac{\mathrm{tr}(\Phi^\top\Phi)}{n} \to \mathrm{tr}(\Sigma)} \cdot \underbrace{\max_{i=1,\dots,n} \mathbb{E}\Big[\varepsilon_i^2 \cdot \mathbf{1}\Big\{\big|\varepsilon_i\big| > \frac{\eta\sqrt{n}}{\big\|\Phi_i\big\|}\Big\}\Big]}_{=:T}.$$

Under condition (ii), $\max_i \frac{\|\Phi_i\|}{\sqrt{n}} \to 0$, so the threshold $\frac{\eta\sqrt{n}}{\|\Phi_i\|} \to \infty$. By the dominated convergence theorem ($\mathbb{E}\big[\varepsilon_i^2\big] = \sigma^2 < \infty$), $T \to 0$. Since the first factor is bounded, the product $\to 0$.

Detail: $T \to 0$. Define $c_{n,i} := \frac{\eta\sqrt{n}}{\|\Phi_i\|} = \frac{\eta}{\|\Phi_i\|/\sqrt{n}}$. By condition (ii), $\max_i \frac{\|\Phi_i\|}{\sqrt{n}} \to 0$, so $\min_{1\le i\le n} c_{n,i} \to \infty$. For each i, since ε_i is a.s. finite, as $c_{n,i} \to \infty$:

$$\varepsilon_i^2 \cdot \mathbf{1}\big\{\big|\varepsilon_i\big| > c_{n,i}\big\} \;\longrightarrow\; 0 \quad \text{a.s.}$$

Moreover, $\varepsilon_i^2 \cdot \mathbf{1}\big\{\big|\varepsilon_i\big| > c_{n,i}\big\} \le \varepsilon_i^2$ with $\mathbb{E}\big[\varepsilon_i^2\big] = \sigma^2 < \infty$, providing an integrable dominating function. By the Dominated Convergence Theorem,

$$\mathbb{E}\big[\varepsilon_i^2 \cdot \mathbf{1}\big\{\big|\varepsilon_i\big| > c_{n,i}\big\}\big] \longrightarrow 0 \quad \text{for each } i.$$

Taking the maximum over $i = 1, \dots, n$ preserves the limit, so $T \to 0$.

Conclusion. By the multivariate Lindeberg–Feller CLT, $\sqrt{n}\,z_n \xrightarrow{d} \mathcal{N}_d(0,\, \sigma^2\Sigma)$. Combined with Step 1 and Slutsky's theorem:

$$\sqrt{n}(\hat\beta - \beta^*) = \hat\Sigma^{-1} \cdot \sqrt{n}\,\frac{\Phi^\top\varepsilon}{n} \xrightarrow{d} \mathcal{N}_d\big(0,\, \sigma^2\Sigma^{-1}\big).$$

10 OLS asymptotic normality in random design

Consistency of $\hat{\beta}$ under random design

Setup. Assume i.i.d. observations $(Y_i, \mathbf{x}_i)$, $i = 1, \dots, n$, from the well-specified linear model $Y_i = \mathbf{x}_i^\top \boldsymbol{\beta} + \varepsilon_i$, where $\mathbf{x}_i \in \mathbb{R}^d$ are random with $\mathbb{E}[\mathbf{x}_i \mathbf{x}_i^\top] = \boldsymbol{\Sigma}$ (positive definite), $E[\varepsilon_i \mid \mathbf{x}_i] = 0$, $\mathrm{Var}(\varepsilon_i \mid \mathbf{x}_i) = \sigma^2$, and $\mathbb{E}[\|\mathbf{x}_i\|^2] < \infty$. We claim $\hat{\boldsymbol{\beta}} \xrightarrow{p} \boldsymbol{\beta}$.

Step 1: Expand $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}$. The OLS estimator is $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$. Substituting $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$:

$$\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\varepsilon} = \big(\frac{\mathbf{X}^\top \mathbf{X}}{n}\big)^{-1} \frac{\mathbf{X}^\top \boldsymbol{\varepsilon}}{n}.$$

Step 2: Show $\frac{\mathbf{X}^\top \mathbf{X}}{n} \xrightarrow{p} \boldsymbol{\Sigma}$. Since $\frac{\mathbf{X}^\top \mathbf{X}}{n} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^\top$ and the $\mathbf{x}_i$ are i.i.d. with finite second moment $\mathbb{E}[\mathbf{x}_i \mathbf{x}_i^\top] = \boldsymbol{\Sigma}$, by the weak law of large numbers:

$$\frac{\mathbf{X}^\top \mathbf{X}}{n} \xrightarrow{p} \boldsymbol{\Sigma}.$$

By the continuous mapping theorem (matrix inversion is continuous at invertible matrices):

$$\big(\frac{\mathbf{X}^\top \mathbf{X}}{n}\big)^{-1} \xrightarrow{p} \boldsymbol{\Sigma}^{-1}.$$

Step 3: Show $\frac{\mathbf{X}^\top \boldsymbol{\varepsilon}}{n} \xrightarrow{p} \mathbf{0}$. Note that $\frac{\mathbf{X}^\top \boldsymbol{\varepsilon}}{n} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \varepsilon_i$. The summands $\mathbf{x}_i \varepsilon_i$ are i.i.d., so it suffices to check their mean and covariance.

Mean. By the law of iterated expectations: $\mathbb{E}[\mathbf{x}_i \varepsilon_i] = \mathbb{E}[\mathbb{E}[\mathbf{x}_i \varepsilon_i \mid \mathbf{x}_i]] = \mathbb{E}[\mathbf{x}_i \, \mathbb{E}[\varepsilon_i \mid \mathbf{x}_i]]$. Since $E[\varepsilon_i \mid \mathbf{x}_i] = 0$:

$$\mathbb{E}[\mathbf{x}_i \varepsilon_i] = \mathbf{0}.$$

Covariance.

$$\mathrm{Cov}(\mathbf{x}_i \varepsilon_i) = \mathbb{E}[\varepsilon_i^2 \, \mathbf{x}_i \mathbf{x}_i^\top] = \mathbb{E}[\mathbb{E}[\varepsilon_i^2 \, \mathbf{x}_i \mathbf{x}_i^\top \mid \mathbf{x}_i]] = \mathbb{E}[\mathbf{x}_i \mathbf{x}_i^\top \, \mathbb{E}[\varepsilon_i^2 \mid \mathbf{x}_i]].$$

Since $E[\varepsilon_i \mid \mathbf{x}_i] = 0$, we have $\mathbb{E}[\varepsilon_i^2 \mid \mathbf{x}_i] = \mathrm{Var}(\varepsilon_i \mid \mathbf{x}_i) = \sigma^2$, so

$$\mathrm{Cov}(\mathbf{x}_i \varepsilon_i) = \sigma^2 \, \mathbb{E}[\mathbf{x}_i \mathbf{x}_i^\top] = \sigma^2 \boldsymbol{\Sigma} < \infty.$$

Since $\mathbf{x}_i \varepsilon_i$ are i.i.d. with mean $\mathbf{0}$ and finite covariance matrix, by the weak law of large numbers:

$$\frac{\mathbf{X}^\top \boldsymbol{\varepsilon}}{n} \xrightarrow{p} \mathbf{0}.$$

Step 4: Conclude via Slutsky's theorem. From Steps 2 and 3, by Slutsky's theorem (product of a sequence converging in probability to a constant and a sequence converging in probability to zero):

$$\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = \big(\frac{\mathbf{X}^\top \mathbf{X}}{n}\big)^{-1} \frac{\mathbf{X}^\top \boldsymbol{\varepsilon}}{n} \xrightarrow{p} \boldsymbol{\Sigma}^{-1} \cdot \mathbf{0} = \mathbf{0}.$$

Hence $\hat{\boldsymbol{\beta}}$ is a consistent estimator of $\boldsymbol{\beta}$.

Asymptotic Normality for $\hat{\beta}$ under random design setting

Setup. Same setup as above. We aim to show:

$$\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) \xrightarrow{d} \mathcal{N}_d\big(0, \sigma^2 \boldsymbol{\Sigma}^{-1}\big).$$

Step 1: Rescale the expansion. From the consistency proof:

$$\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) = \big(\frac{\mathbf{X}^\top \mathbf{X}}{n}\big)^{-1} \frac{\mathbf{X}^\top \boldsymbol{\varepsilon}}{\sqrt{n}}.$$

Step 2: Apply the multivariate CLT to $\frac{\mathbf{X}^\top \boldsymbol{\varepsilon}}{\sqrt{n}}$. We have $\frac{\mathbf{X}^\top \boldsymbol{\varepsilon}}{\sqrt{n}} = \frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{x}_i \varepsilon_i$. From the consistency proof, $\mathbf{x}_i \varepsilon_i$ are i.i.d. with $\mathbb{E}\big[\mathbf{x}_i \varepsilon_i\big] = \mathbf{0}$ and $\operatorname{Cov}(\mathbf{x}_i \varepsilon_i) = \sigma^2 \boldsymbol{\Sigma}$. By the multivariate CLT:

$$\frac{\mathbf{X}^\top \boldsymbol{\varepsilon}}{\sqrt{n}} \xrightarrow{d} \mathcal{N}_d\big(\mathbf{0}, \, \sigma^2 \boldsymbol{\Sigma}\big).$$

Step 3: Conclude via Slutsky's theorem. We already know $\big(\frac{\mathbf{X}^\top \mathbf{X}}{n}\big)^{-1} \xrightarrow{p} \boldsymbol{\Sigma}^{-1}$ (a constant matrix). By Slutsky's theorem (product of a sequence converging in probability to a constant and a sequence converging in distribution):

$$\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) = \big(\frac{\mathbf{X}^\top \mathbf{X}}{n}\big)^{-1} \frac{\mathbf{X}^\top \boldsymbol{\varepsilon}}{\sqrt{n}} \xrightarrow{d} \mathcal{N}_d\big(\mathbf{0}, \, \sigma^2 \boldsymbol{\Sigma}^{-1}\big).$$

The asymptotic covariance $\sigma^2 \boldsymbol{\Sigma}^{-1}$ follows from $\boldsymbol{\Sigma}^{-1}(\sigma^2 \boldsymbol{\Sigma})(\boldsymbol{\Sigma}^{-1})^\top = \sigma^2 \boldsymbol{\Sigma}^{-1} \boldsymbol{\Sigma} \, \boldsymbol{\Sigma}^{-1} = \sigma^2 \boldsymbol{\Sigma}^{-1}$.

11 Assumption lean regression

Assumption lean setting: Assume the pair $(Y, X) \sim P_{Y,X}$ on $\mathbb{R} \times \mathbb{R}^d$ have each 2 moments: $\mathbb{E}\big[Y^2\big] < \infty$ and $\Sigma = \mathbb{E}\big[XX^\top\big]$ is invertible.
Then the projection parameter β^* is well defined as:

$$\begin{aligned}
\beta^* &= \underset{\beta \in \mathbb{R}^d}{\operatorname{argmin}} \, \mathbb{E}\Big[\big(Y - X^\top \beta\big)^2\Big] \\
&= \underset{\beta \in \mathbb{R}^d}{\operatorname{argmin}} \, \mathbb{E}\Big[\big(\mathbb{E}\big[Y \mid X\big] - X^\top \beta\big)^2\Big] \\
&= \Sigma^{-1} \underbrace{\mathbb{E}\big[Y \cdot X\big]}_{\Gamma \in \mathbb{R}^d}
\end{aligned}$$

Remarks

Remark 1. Based on the above derivation, we obtain the population-level normal equations: $\mathbb{E}\big[XX^\top\big]\beta^* = \mathbb{E}\big[XY\big]$. The corresponding sample-level normal equations: $X^\top X\hat{\beta} = X^\top y$. Note, X is a random vector at the population-level and a deterministic matrix $\in \mathbb{R}^{n\times d}$ at the sample-level.
Remark 2. Using the theory of L_2 projections, the population level normal equation implies that

$$\mathbb{E}\Big[\big(Y - X^\top \beta^*\big)\, X^\top a\Big] = \mathbb{E}\Big[\big(\mathbb{E}\big[Y \mid X\big] - X^\top \beta^*\big)\, X^\top a\Big] = 0 \quad \forall\, a \in \mathbb{R}^d$$

The two orthogonality conditions in two different L^2 spaces:

- **Left-hand side:** This is a projection of Y in the full space $L^2(\Omega, P)$. The residual $Y - X^\top\beta^*$ contains both the non-linearity $\eta = \mathbb{E}[Y \mid X] - X^\top\beta^*$ and the pure noise $\varepsilon = Y - \mathbb{E}[Y \mid X]$.
- **Right-hand side:** This is a projection of $\mathbb{E}[Y \mid X]$ in the smaller space $L^2(P_X)$, i.e., the space of square-integrable functions of X alone. The noise ε has been eliminated by the conditional expectation, and the residual is purely the non-linear component η.

Geometrically, β^* is simultaneously the best linear approximation in two different L^2 spaces, and the tower property is precisely what connects them.
Remark 3. It can be interpreted from a **projection** perspective, since β^* is a projection parameter, and we have $\mathbb{E}\big[X(Y - X^\top\beta^*)\big] = 0$. The corresponding sample-level projection formula is: $X^\top(y - X^\top\hat{\beta}) = 0$.
Remark 4. It can also be interpreted from a **M-estimation** perspective, since $\mathbb{E}\big[X(Y - X^\top\beta^*)\big] = 0$ is a estimating equation. The corresponding sample-level estimating equation is: $\frac{1}{n}\sum_{i=1}^n [X_i(y_i - X_i^\top\hat{\beta})] = 0$.
Remark 5. $\varepsilon = \big(Y - \mathbb{E}\big[Y \mid X\big]\big)$ is orthogonal to all r.v.'s of the form $f(X)$, any f s.t. $\mathrm{Var}\big[f(X)\big] < \infty$, i.e. $\mathbb{E}\big[\varepsilon \cdot f\big] = 0$.
Remark 6. When the model is misspecified, the projection parameter $\beta^* = \Sigma^{-1}\mathbb{E}\big[XY\big] = \big(\mathbb{E}\big[XX^\top\big]\big)^{-1}\mathbb{E}\big[XY\big]$ depends on the marginal distribution P_X, since both Σ and $\mathbb{E}\big[XY\big]$ involve P_X. Changing the distribution of X changes β^*. When the model is well-specified, i.e. $\mathbb{E}\big[Y \mid X\big] = X^\top\beta^*$, β^* is determined entirely by the conditional distribution $P_{Y|X}$ and does not depend on P_X.

good luck :)

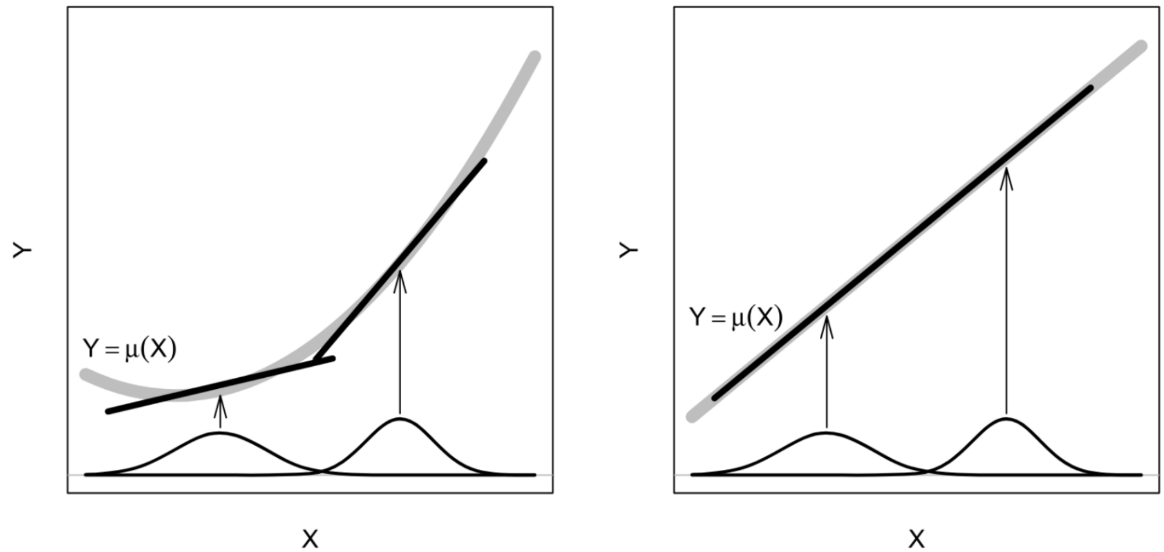

Figure 1: Illustration of the dependence of the population OLS solution on the marginal distribution of the regressors: The left figure shows dependence in the presence of nonlinearity; the right figure shows independence in the presence of linearity.

Remark 7. Building on Remark 6 and Figure 1, the role of X relative to β^* differs fundamentally between the two specification regimes:

(a) Well-specified. When $\mathbb{E}\big[Y \mid X\big] = X^\top \beta^*$, the design matrix X is an *ancillary statistic* for β^*: its distribution P_X carries no information about the parameter. This has three immediate consequences. First, β^* is determined entirely by $P_{Y|X}$ and is invariant to changes in P_X. Second, the OLS estimator $\hat{\beta} = \hat{\Sigma}^{-1}\hat{\Gamma}$ is unbiased unconditionally, since $\mathbb{E}\big[\hat{\beta} \mid X\big] = \beta^*$ is constant in X, so by the law of total expectation $\mathbb{E}\big[\hat{\beta}\big] = \beta^*$. Third, in the total-variance decomposition

$$\operatorname{Var}\big[\hat{\beta}\big] = \mathbb{E}\big[\operatorname{Var}\big[\hat{\beta} \mid X_1, \ldots, X_n\big]\big] + \operatorname{Var}\big[\mathbb{E}\big[\hat{\beta} \mid X_1, \ldots, X_n\big]\big],$$

the second term vanishes because $\mathbb{E}\big[\hat{\beta} \mid X\big] = \beta^*$ does not vary with X.

(b) Misspecified. When $\mathbb{E}\big[Y \mid X\big] \neq X^\top \beta$ for any β, the projection parameter $\beta^* = \big(\mathbb{E}\big[XX^\top\big]\big)^{-1}\mathbb{E}\big[XY\big]$ depends on P_X: intuitively, P_X determines where along the nonlinear curve $\mu(X)$ the linear approximation is weighted most heavily (Figure 1, left panel). Thus X is *no longer ancillary.* This has two consequences. First, $\mathbb{E}\big[\hat{\beta} \mid X\big]$ is generally not equal to β^*, and $\mathbb{E}\big[A^{-1}B\big] \neq \big(\mathbb{E}\big[A\big]\big)^{-1}\mathbb{E}\big[B\big]$ prevents the expectation from passing through the ratio $\hat{\Sigma}^{-1}\hat{\Gamma}$, so $\hat{\beta}$ is generally biased. Second, the term $\operatorname{Var}\big[\mathbb{E}\big[\hat{\beta} \mid X\big]\big]$ in the total-variance decomposition is nonzero, introducing an additional source of variability driven by the randomness in the design.

In short, well-specification makes P_X irrelevant to both the *target* β^* and the *estimation quality* of $\hat{\beta}$; misspecification entangles both with P_X, which is especially problematic under covariate shift where the training and target distributions of X differ.

Consistentency of $\hat{\beta}$ under assumption lean setting: $\hat{\beta}$ is nonetheless a consistent estimator of β^*: $\hat{\beta} \xrightarrow{p} \beta^*$. The proof idea is: using the law of large numbers and Slutsky's theorem.
Motivation behind assumption lean setting CLT for $\hat{\beta}$: When the model is misspecified, the residual variance is no longer constant. Note $\eta = \mathbb{E}\big[Y \mid X\big] - X^\top \beta^*$ is a deterministic function of X, so $\mathrm{Var}(\eta \mid X) = 0$. Thus

$$\mathrm{Var}(Y - X^\top \beta^* \mid X) = \mathrm{Var}(\eta + \varepsilon \mid X) = \mathrm{Var}(\varepsilon \mid X),$$

which is a function of X (heteroscedasticity), not a constant.
The classical OLS variance formula assumes homoscedasticity:

$$\mathrm{Var}(\hat{\beta} \mid X) = \sigma^2 (X^\top X)^{-1}, \quad \sigma^2 = \mathrm{Var}(\varepsilon \mid X) = \text{const}.$$

Under heteroscedasticity, the correct conditional variance is the sandwich form:

$$\mathrm{Var}(\hat{\beta} \mid X) = (X^\top X)^{-1} \Big(\sum_{i=1}^{n} X_i X_i^\top \sigma^2(X_i) \Big) (X^\top X)^{-1},$$

where $\sigma^2(X_i) = \mathrm{Var}(\varepsilon \mid X = X_i)$ varies across observations. The sandwich variance $\Sigma^{-1} V \Sigma^{-1}$ must replace the classical $\sigma^2 \Sigma^{-1}$, because the variance itself is a function of X, not a constant. Heteroscedasticity and model misspecification are two independent issues: a well-specified model can still have heteroscedastic errors, and a misspecified model always requires worrying about it. Always use the sandwich standard error.
OLS estimator $\hat{\beta}$'s bias and variance analysis under assumption lean setting (random design): Assume n i.i.d. pairs (Y_i, X_i), $i = 1, \ldots, n$, from the unknown data generating distribution $P_{Y,X}$. We estimate β^* using OLS:

$$\hat{\beta} = \hat{\Sigma}^{-1} \hat{\Gamma}, \quad \text{where} \quad \hat{\Sigma} = \frac{1}{n} \sum_{i=1}^{n} X_i X_i^\top, \quad \hat{\Gamma} = \frac{1}{n} \sum_{i=1}^{n} Y_i X_i$$

Then $\mathbb{E}\big[\hat{\beta}\big] \neq \beta^*$ in general. This is because $\hat{\beta} = \hat{\Sigma}^{-1} \hat{\Gamma}$ is a ratio of two random quantities, and in general $\mathbb{E}\big[A^{-1} B\big] \neq \big(\mathbb{E}\big[A\big]\big)^{-1} \mathbb{E}\big[B\big]$. $\mathbb{E}\big[\hat{\beta}\big] \neq \beta^*$ if the model is well-specified. The proof of unbiased-ness well specification is based on the law of total expectation. under By the law of total variance:

$$\mathrm{Var}\big[\hat{\beta}\big] = \mathbb{E}\big[\mathrm{Var}\big[\hat{\beta} \mid X_1, \ldots, X_n\big]\big] + \underbrace{\mathrm{Var}\big[\mathbb{E}\big[\hat{\beta} \mid X_1, \ldots, X_n\big]\big]}_{=0 \text{ when well-specified}}$$

CLT for $\hat{\beta}$ under assumption lean setting
Step 1: Define the influence function ψ_i. To establish a CLT for $\hat{\beta}$, consider the following quantities (not computable since β^* is unknown):

$$\psi_i := \Sigma^{-1} X_i (Y_i - X_i^\top \beta^*), \quad i = 1, \ldots, n.$$

Step 2: Moments of ψ_i.

- $\mathbb{E}\big[\psi_i\big] = \Sigma^{-1} \mathbb{E}\big[X (Y - X^\top \beta^*)\big] = 0$ (by the normal equations).
- $\mathrm{Var}(\psi_i) = \Sigma^{-1} V \Sigma^{-1}$, where $V := \mathrm{Var}\big(X_i (Y_i - X_i^\top \beta^*)\big)$ is the sandwich variance meat.

Derivation of $\mathrm{Var}(\psi_i)$: since $\mathbb{E}\big[\psi_i\big] = 0$, $\mathrm{Var}(\psi_i) = \mathbb{E}\big[\psi_i \psi_i^\top\big] = \Sigma^{-1} \underbrace{\mathbb{E}\big[X_i X_i^\top (Y_i - X_i^\top \beta^*)^2\big]}_{=V} \Sigma^{-1}$, using $\mathrm{Var}(AZ) = A\,\mathrm{Var}(Z) A^\top$ with $A = \Sigma^{-1}$. (When model is well-specified and homoscedastic: $V = \sigma^2 \Sigma$, so $\mathrm{Var}(\psi_i) = \sigma^2 \Sigma^{-1}$, recovering the classical result.)
Step 3: Multivariate CLT. Since ψ_i are i.i.d. with mean 0 and covariance $\Sigma^{-1} V \Sigma^{-1}$, by the multivariate CLT:

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{n} \psi_i \xrightarrow{d} N_d\big(0,\ \Sigma^{-1} V \Sigma^{-1}\big).$$

Step 4: Key algebra — linking $\sqrt{n}(\hat{\beta} - \beta^*)$ to $\frac{1}{\sqrt{n}} \sum \psi_i$. Recall $\hat{\Sigma} = \frac{1}{n} \sum_i X_i X_i^\top$ and $\hat{\Gamma} = \frac{1}{n} \sum_i X_i Y_i$. From the sample normal equations $\hat{\Sigma} \hat{\beta} = \hat{\Gamma}$:

$$\hat{\Sigma}(\hat{\beta} - \beta^*) = \hat{\Gamma} - \hat{\Sigma} \beta^*.$$

Averaging ψ_i and substituting:

$$\frac{1}{n}\sum_{i=1}^{n}\psi_i = \Sigma^{-1}(\hat{\Gamma}-\hat{\Sigma}\beta^*) = \Sigma^{-1}\hat{\Sigma}(\hat{\beta}-\beta^*).$$

Multiplying both sides by $\sqrt{n}$:

$$\Sigma^{-1}\hat{\Sigma}\cdot\sqrt{n}(\hat{\beta}-\beta^*) = \frac{1}{\sqrt{n}}\sum_{i=1}^{n}\psi_i.$$

Step 5: Conclude via CMT and Slutsky. Since $\hat{\Sigma}\xrightarrow{p}\Sigma$, by CMT: $(\Sigma^{-1}\hat{\Sigma}-I_d)\xrightarrow{p}0$, hence

$$\Sigma^{-1}\hat{\Sigma}\cdot\sqrt{n}(\hat{\beta}-\beta^*) - \sqrt{n}(\hat{\beta}-\beta^*) = o_p(1).$$

Combining with Step 3 and Step 4, by Slutsky:

$$\sqrt{n}(\hat{\beta}-\beta^*)\xrightarrow{d}N_d\big(0,\ \Sigma^{-1}V\Sigma^{-1}\big).$$

Remark. If the model is well-specified and $Y_i - X_i^\top\beta^* = \varepsilon_i \sim (0,\sigma^2)$, then

$$\mathrm{Var}\big(\psi_i\big) = \mathrm{Var}\big(\Sigma^{-1}X_i(Y_i - X_i^\top\beta^*)\big) = \Sigma^{-1}\underbrace{\mathbb{E}\big[X_iX_i^\top\varepsilon_i^2\big]}_{=\sigma^2\Sigma}\Sigma^{-1} = \sigma^2\Sigma^{-1},$$

recovering the classical OLS variance formula.

Proof that $\hat{V}\xrightarrow{p}V$.

Step 1: Introduce intermediate quantity $\tilde{V}$. Define (not computable since β^* unknown):

$$\tilde{V} = \frac{1}{n}\sum_{i=1}^{n}X_iX_i^\top(Y_i - X_i^\top\beta^*)^2 \xrightarrow{p} V \quad \text{(WLLN)}.$$

It suffices to show $\hat{V}-\tilde{V}\xrightarrow{p}0$.

Step 2: Expand $\hat{V}-\tilde{V}$. We have:

$$\hat{V}-\tilde{V} = \frac{1}{n}\sum_{i=1}^{n}X_iX_i^\top\Big[(X_i^\top\hat{\beta})^2 - (X_i^\top\beta^*)^2 + 2Y_iX_i^\top(\beta^*-\hat{\beta})\Big].$$

Let $\delta = \hat{\beta}-\beta^*$. Expand $(X_i^\top\hat{\beta})^2 = (X_i^\top(\beta^*+\delta))^2 = (X_i^\top\beta^*)^2 + 2(X_i^\top\beta^*)(X_i^\top\delta) + (X_i^\top\delta)^2$, so $(X_i^\top\hat{\beta})^2 - (X_i^\top\beta^*)^2 = 2(X_i^\top\beta^*)(X_i^\top\delta) + (X_i^\top\delta)^2$. Substituting and combining with the $2Y_iX_i^\top(\beta^*-\hat{\beta})$ term:

$$\hat{V}-\tilde{V} = \frac{1}{n}\sum_{i=1}^{n}X_iX_i^\top\Big[\underbrace{(X_i^\top(\hat{\beta}-\beta^*))^2}_{\text{quadratic term}} + \underbrace{2(Y_i - X_i^\top\beta^*)X_i^\top(\hat{\beta}-\beta^*)}_{\text{cross term}}\Big].$$

Step 3: Op-norm bound via Cauchy–Schwarz. For any $M = \frac{1}{n}\sum_i X_iX_i^\top c_i$, since $\big\|X_iX_i^\top\big\|_{\mathrm{op}} = \big\|X_i\big\|^2$ (rank-one matrix), $\big\|M\big\|_{\mathrm{op}} \le \frac{1}{n}\sum_i\big\|X_i\big\|^2|c_i|$. Applying to both terms:

$$\big\|\hat{V}-\tilde{V}\big\|_{\mathrm{op}} \le \frac{1}{n}\sum_i\big\|X_i\big\|^2(X_i^\top(\hat{\beta}-\beta^*))^2 + \frac{2}{n}\sum_i\big\|X_i\big\|^2\big|Y_i - X_i^\top\beta^*\big|\cdot\big|X_i^\top(\hat{\beta}-\beta^*)\big|.$$

Apply C-S ($\sum a_ib_i \le \sqrt{\sum a_i^2}\sqrt{\sum b_i^2}$) to the cross term with $a_i = \big\|X_i\big\|\big|X_i^\top(\hat{\beta}-\beta^*)\big|$, $b_i = \big\|X_i\big\|\big|Y_i - X_i^\top\beta^*\big|$:

$$\big\|\hat{V}-\tilde{V}\big\|_{\mathrm{op}} \le \underbrace{\frac{1}{n}\sum_i\big\|X_i\big\|^2(X_i^\top(\hat{\beta}-\beta^*))^2}_{=:A} + 2\sqrt{A}\cdot\sqrt{\underbrace{\frac{1}{n}\sum_i\big\|X_i\big\|^2(Y_i - X_i^\top\beta^*)^2}_{=:B}} = A + 2\sqrt{A}\sqrt{B}.$$

Step 4: Show $A \xrightarrow{p} 0$. By C-S $(X_i^\top(\hat\beta - \beta^*))^2 \le \|X_i\|^2 \|\hat\beta - \beta^*\|^2$:

$$A \le \underbrace{\|\hat\beta - \beta^*\|^2}_{\xrightarrow{p} 0} \cdot \underbrace{\frac{1}{n}\sum_i \|X_i\|^4}_{\xrightarrow{p} \mathbb{E}[\|X\|^4] < \infty} \xrightarrow{p} 0. \quad \text{(Slutsky)}$$

Step 5: Show $\sqrt{B} = O_p(1)$. By WLLN: $B \xrightarrow{p} \mathbb{E}\big[\|X\|^2 (Y - X^\top\beta^*)^2\big]$. Use trace trick $\|X\|^2 = \mathrm{tr}(XX^\top)$ and linearity of trace and expectation:

$$\mathbb{E}\big[\|X\|^2 (Y - X^\top\beta^*)^2\big] = \mathrm{tr}\big(\mathbb{E}\big[XX^\top (Y - X^\top\beta^*)^2\big]\big) = \mathrm{tr}(V) < \infty,$$

so $B \xrightarrow{p} \mathrm{tr}(V) < \infty$, hence $\sqrt{B} = O_p(1)$.
Step 6: Conclude by Slutsky. $A \xrightarrow{p} 0$ and $\sqrt{B} = O_p(1)$ give $A + 2\sqrt{A}\sqrt{B} \xrightarrow{p} 0$, so $\hat V - \tilde V \xrightarrow{p} 0$. With $\tilde V \xrightarrow{p} V$: $\hat V \xrightarrow{p} V$. Combined with $\hat\Sigma^{-1} \xrightarrow{p} \Sigma^{-1}$ (CMT):

$$\sqrt{n}(\hat\beta - \beta^*) \approx N_d\big(0,\ \hat\Sigma^{-1}\hat V \hat\Sigma^{-1}\big).$$